EE 3072
System Dynamics and Control

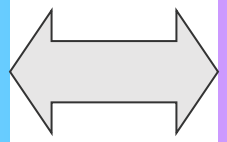
The Root-locus Technique

Outline

- Introduction
- Root-locus equation
- Rules to draw regular root loci
- Generalized root loci
- Analysis of control system by the root-locus method
- Control system design by the root-locus method

Introduction

Stability and the characteristics of transient response of closed-loop systems

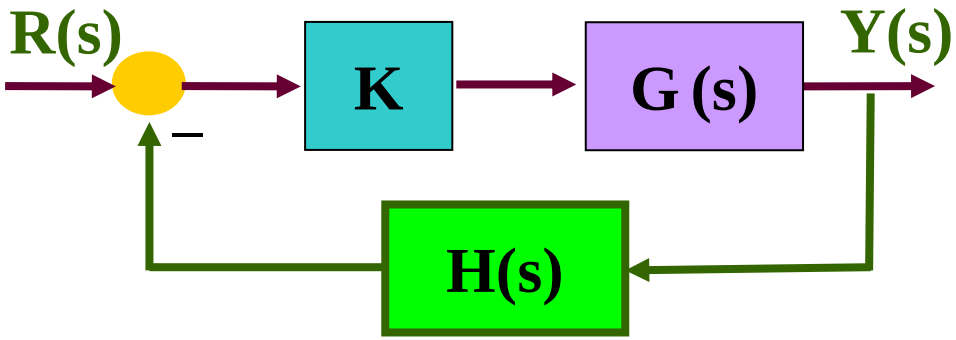


Locations of the closed-loop poles

Problems in calculating the closed loop poles (i.e., the roots of the closed loop characteristic equation):

1. Difficult for a system of third or higher order.
2. Tedious for varying parameters.

Varying the loop gain K



The open loop gain K is an important parameter that can affect the performance of a system

- In many systems, simple gain adjustment may move the closed-loop poles to desired locations.
- Then the design problem may become the selection of an appropriate gain value.
- It is important to know how the closed-loop poles move in the s plane as the loop gain K is varied.

Root locus method

A simple method for finding the roots of the characteristic equation has been developed by W. R. Evans. This method is called root locus method.

- Graphical Analysis of Control System, AIEE Trans. Part II,67(1948),pp.547-551.
- Control System Synthesis by Root Locus Method, AIEE Trans. Part II,69(1950),pp.66-69

Root Locus:

The locus of roots of the characteristic equation of the closed-loop system as a specific parameter (usually, gain K) is varied form 0 to ∞.

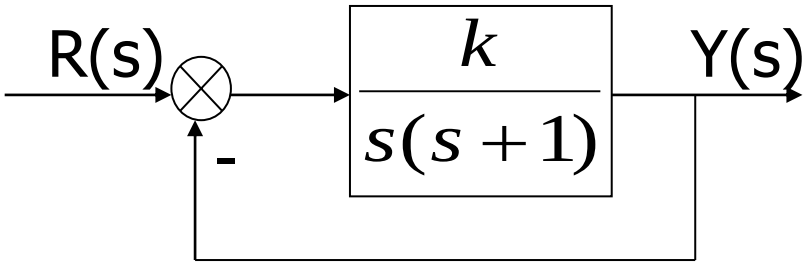
The advantages of root locus approach:

1. Avoiding tedious and complex roots-solving calculation
2. Clearly showing the contributions of the loop poles and/or zeros to the location of the closed-loop poles.
3. Indicating the manner in which the loop poles and zeros should be modified so that the response meets system performance specifications.

Root-locus equation

Start from an example

Consider a second-order system shown as follows:



Closed-loop TF:

$$T(s)=\frac{k}{s^2+s+k}$$

Characteristic equation (CE): $s^2+s+k=0$

Roots of CE: $$s_{1,2}=\frac{-1\pm\sqrt{1-4k}}{2}=-\frac{1}{2}\pm\frac{1}{2}\sqrt{1-4k}$$

The roots of CE change as the value of k changes.

When k changes from 0 to ∞, how will the locus of the roots of CE move?

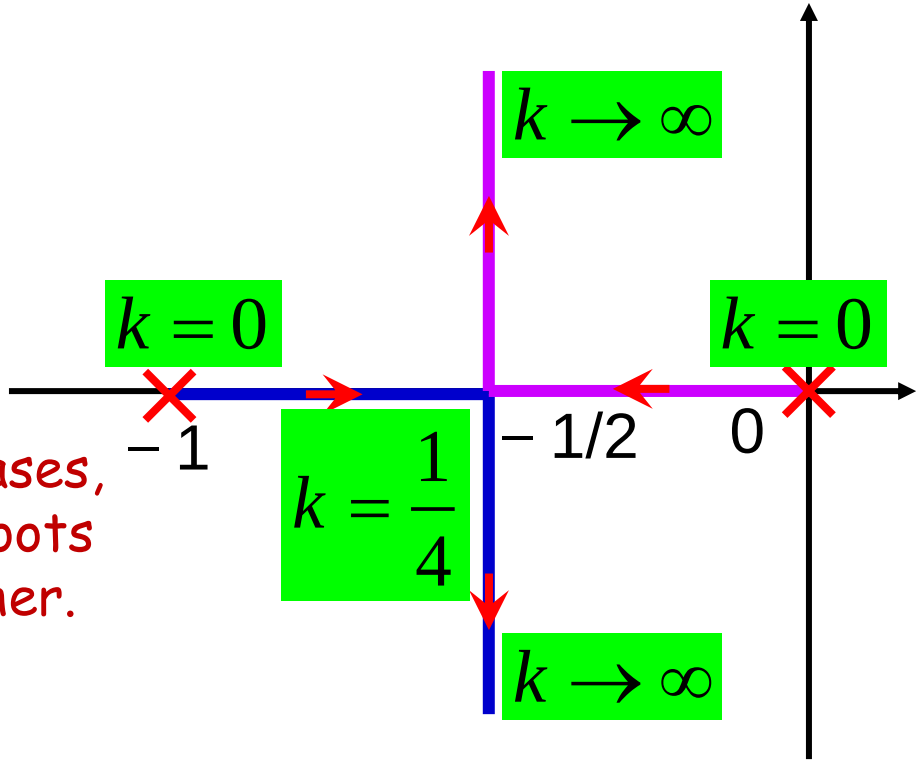
$$s_{1,2} = -\frac{1}{2} \pm \frac{1}{2}\sqrt{1-4k}, \quad k: 0 \to \infty$$

$k = 0$ $\quad s_1 = 0 \quad s_2 = -1$

$0 < k < 1/4$ As the value of k increases, the two negative real roots move closer to each other.

$k = 1/4$ $\quad s_1 = s_2 = -1/2$

$1/4 < k < \infty$ A pair of complex-conjugate roots leave the negative real-axis and move upwards and downwards following the line s=-1/2.



On the s plane, using arrows to denote the direction of characteristic roots move when k increases, by numerical value to denote the gain at the poles.

By Root loci, we can analyze the system behaviors

(1) Stability:

when Root loci are on the left half plane, then the system is definitely stable for all k > 0.

(2) Steady-state performance:

there is an open-loop pole at s = 0, so the system is a type I system. The steady-state error is

0	under step input signal
v_0/K_v	under ramp signal $v_0 t$
∞	under parabolic signal.

(3) Transient performance:

there's a close relationship between root loci and system behavior

on the real-axis: $k < 0.25$ overdamped;

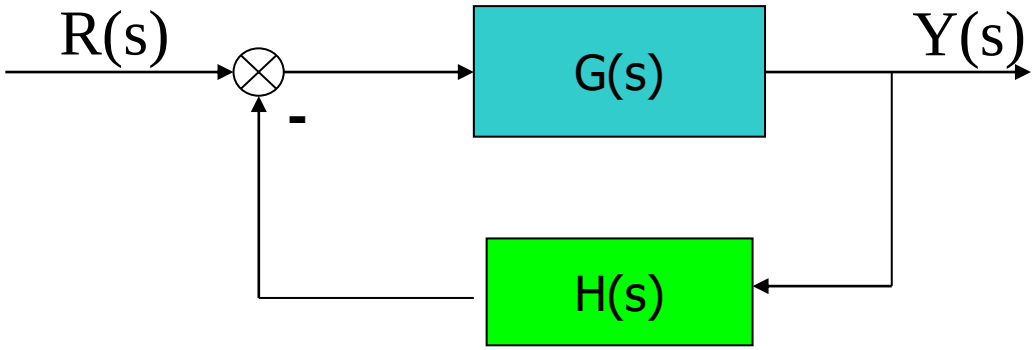
$k = 0.25$ critically damped

$k > 0.25$ underdamped.

However, in general it's difficult to draw root loci directly by closed-loop characteristic roots-solving method.

The idea of root locus method: using loop transfer function, draw closed-loop root loci directly.

Relationship between zeros and poles of G(s)H(s) and closed-loop ones



Forward path TF: $G(s)$

Feedback path TF: $H(s)$

Loop TF: $G(s)H(s)$

Closed-loop TF:

$$T(s) = \frac{G(s)}{1 + G(s)H(s)}$$

Write G(s) and H(s) into zero-pole-gain (zpk) form:

$$G(s) = \frac{K_G \prod_{i=1}^{a} (s - Z_i)}{\prod_{i=1}^{b} (s - P_i)} \qquad H(s) = \frac{K_H \prod_{j=1}^{c} (s - Z_j)}{\prod_{j=1}^{d} (s - P_j)}$$

K_G ⟶ Gain/coefficient of forward path

K_H ⟶ Gain/coefficient of feedback path

Z_i ⟶ Zeros of forward path TF

P_i ⟶ Poles of forward path TF

Z_j ⟶ Zeros of feedback path TF

P_j ⟶ Poles of feedback path TF

$$G(s)H(s)=\frac{K\prod_{i=1}^{a}(s-Z_i)\prod_{j=1}^{c}(s-Z_j)}{\prod_{i=1}^{b}(s-P_i)\prod_{j=1}^{d}(s-P_j)},$$

Loop gain:

$$K=K_GK_H$$

$$T(s)=\frac{G(s)}{1+G(s)H(s)}=\frac{K_G\prod_{i=1}^{a}(s-Z_i)\prod_{j=1}^{d}(s-P_j)}{\prod_{i=1}^{b}(s-P_i)\prod_{j=1}^{d}(s-P_j)+K\prod_{i=1}^{a}(s-Z_i)\prod_{j=1}^{c}(s-Z_j)}$$

1. The closed-loop zeros = feed forward path zeros + feedback path poles.
2. The closed-loop poles are related to poles and zeros of G(s)H(s) and loop gain K.
3. The closed-loop gain = the feed forward path root loci gain K_G .

Root-locus equation

To draw closed-loop root-locus is to solve the CE

$$1+G(s)H(s)=0$$

That is

$$G(s)H(s)=-1$$ Root-locus equation

Suppose that G(s)H(s) has m zeros (Z_i) and n poles (P_i), the above equation can be re-written as

$$G(s)H(s)=\frac{K\prod_{i=1}^{m}(s-Z_i)}{\prod_{i=1}^{n}(s-P_i)}=-1$$

K varies from 0 to ∞

Root-locus equation:

$$G(s)H(s) = \frac{K\prod_{i=1}^{m}(s - Z_i)}{\prod_{i=1}^{n}(s - P_i)} = -1$$

Since *G(s)H(s)* is function of the complex variable s, the root-locus equation can be described by the following two equations:

Magnitude equation (ME)

$$\frac{K\prod_{i=1}^{m}|s - Z_i|}{\prod_{i=1}^{n}|s - P_i|} = 1$$

Angle equation (AE)

$$\sum_{i=1}^{m}\angle(s - Z_i) - \sum_{i=1}^{n}\angle(s - P_i) = (2l+1)\pi, \quad l = 0, \pm1, \pm2, \cdots$$

Magnitude equation is related not only to zeros and poles of G(s)H(s), but also to loop gain K.

Angle equation is only related to zeros and poles of G(s)H(s).

Use AE to draw root loci and use ME to determine the value of K on root loci.

Example 1

$$G(s)H(s)=\frac{K(\tau s+1)}{s(T_1 s+1)(T_2^2 s^2+2\zeta T_2 s+1)}$$

$$=\frac{K'(s+1/\tau)}{s(s+1/T_1)(s+\zeta\omega_n+j\omega_d)(s+\zeta\omega_n-j\omega_d)}$$

$$K'=K\tau/T_1T_2^2$$

$$\omega_d=\sqrt{1-\zeta^2}\,/T_2$$

$$\omega_n=1/T_2$$

Poles of *G(s)H(s)* (×) $p_1=0$ $p_2=-1/T_1$ $p_{3,4}=-\zeta\omega_n\pm j\omega_d$

Zeros of *G(s)H(s)* (○) $z_1=-1/\tau$

For a point s_1 to be on the root loci

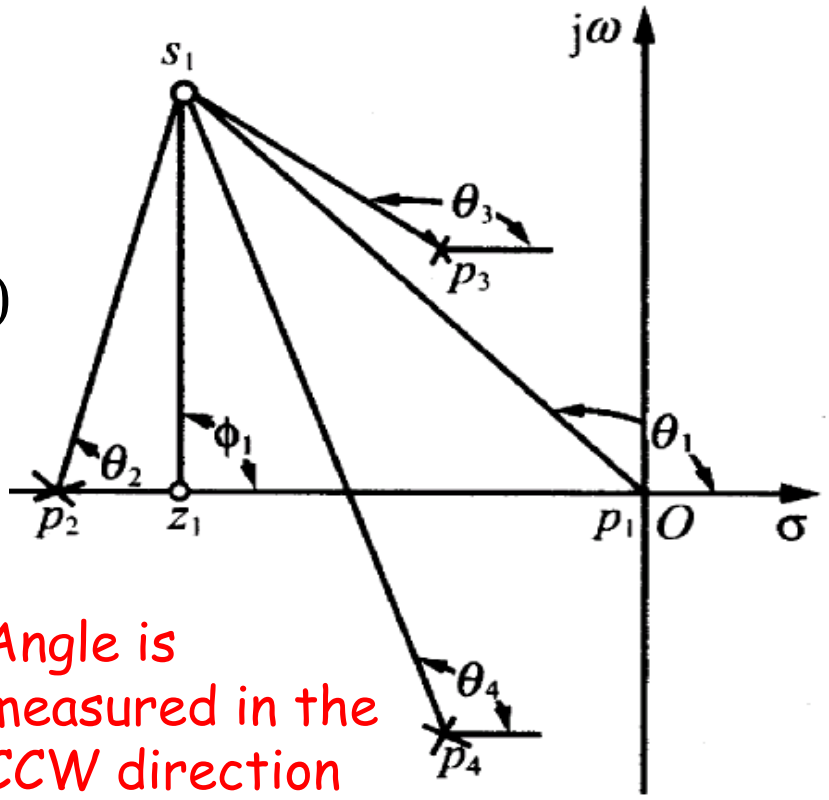
$$\angle G(s_1)H(s_1)=\sum_{i=1}^{1}\angle(s_1-z_i)-\sum_{i=1}^{4}\angle(s_1-p_i)$$

$$=\phi_1-(\theta_1+\theta_2+\theta_3+\theta_4)$$

$$=(2l+1)\pi$$

$$K'=\frac{|s_1|\cdot|s_1-p_2|\cdot|s_1-p_3|\cdot|s_1-p_4|}{|s_1-z_1|}$$

must be satisfied.



Angle is measured in the CCW direction wrt the real axis

Example 2

Unity-feedback transfer function: $G(s) = \dfrac{K}{s}$

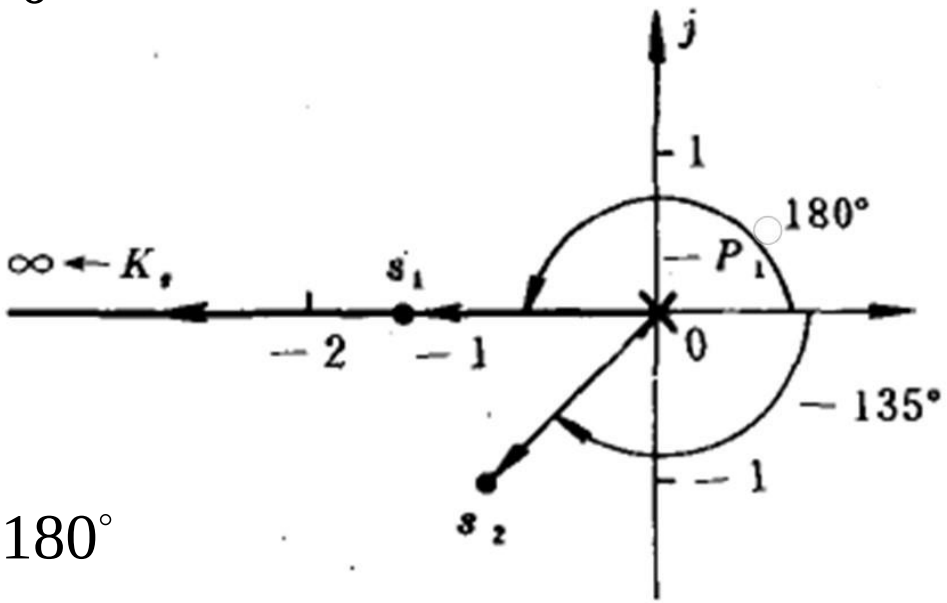
One pole of G(s)H(s): $p_1 = 0$

No zero.

Test a point s_1 on the negative real-axis

$$\sum_{i=1}^{m} \angle(s - z_i) - \sum_{i=1}^{n} \angle(s - p_i)\Big|_{s_1}$$

$$= -\angle(s_1 - p_1) = -\angle s_1 = -180^\circ$$



All the points on the negative real-axis are on the locus.

Test a point outside the negative real-axis s_2=-1-j

$$\sum_{i=1}^{m} \angle(s - z_i) - \sum_{i=1}^{n} \angle(s - p_i)\Big|_{s_2}$$

$$= -\angle(s_2 - p_1) = -\angle s_2 = 135^\circ$$

All the points outside the negative real-axis are not on locus.

Probe by each point:

1. Find all the points that satisfy the Angle Equation on the s-plane, and then link all these points into a smooth curve, thus we have the system root locus when K changes from 0 to ∞;
2. As for the given K, find the points that satisfy the Magnitude Equation on the root locus, then these points are the required closed-loop poles.

However, it's unrealistic to apply such "probe by each point" method.

W.R. Evans (1948) proposed a set of root loci drawing rules which simplify the drawing work.

Rules to draw regular root loci

(suppose the varying parameter is open-loop gain K)

Properties of Root Loci

1 $K = 0$ and $K = \pm\infty$ points of Root Loci

2 Number of Branches on the RL

3 Symmetry of the RL

4 Root Loci on the real-axis

5 Asymptotes of the RL

6 Breakaway (or break-in) points on the RL

7 Departure angle and arrival angle of RL

8 Intersection of the RL with the imaginary axis

9 The sum of the roots and the product of the roots of the closed-loop characteristic equation

1 $K=0$ and $K=\infty$ points

Root loci originate on the poles of G(s)H(s) (for K=0) and terminates on the zeros of G(s)H(s) (as K=∞).

RL Equation:

$$G(s)H(s)=\frac{K\prod_{i=1}^{m}(s-Z_i)}{\prod_{i=1}^{n}(s-P_i)}=-1$$

Magnitude Equation:

$$\frac{K\prod_{i=1}^{m}|s-Z_i|}{\prod_{i=1}^{n}|s-P_i|}=1 \quad \Rightarrow \quad K=\frac{\prod_{i=1}^{n}|s-P_i|}{\prod_{i=1}^{m}|s-Z_i|}$$

$K=0 \quad \rightarrow s=P_i$ →Root loci start from poles of G(s)H(s)

$K=\infty \quad \rightarrow s=Z_i$ →Root loci end at zeros of G(s)H(s).

2 Number of branches on the RL

RL Equation:

$$G(s)H(s) = \frac{K\prod_{i=1}^{m}(s - Z_i)}{\prod_{i=1}^{n}(s - P_i)} = -1$$

The order of the characteristic equation is max(n,m)

→ as K varies from 0 to ∞, max(n,m) roots change

→ max(n,m) root loci

→ max(n,m) branches.

RL have max(n,m) root loci, i.e., number of branches

2 Number of branches on the RL

n'th-order system, RL have n starting points and RL have n branches

RL Equation:

$$G(s)H(s) = \frac{K\prod_{i=1}^{m}(s - Z_i)}{\prod_{i=1}^{n}(s - P_i)} = -1$$

For a real physical system, usually the number of poles of G(s)H(s) is more than the number of its zeros, i.e. n > m.

n root loci in the system – n branches;

all n branches start at the open loop poles;

m root loci (branches) end at open-loop zeros (finite zeros);

$(n-m)$ root loci (branches) end at $(n-m)$ infinite zeros.

2 Number of branches on the RL

m'th-order system, RL have m starting points and RL have m branches

RL Equation:

$$G(s)H(s) = \frac{K\prod_{i=1}^{m}(s - Z_i)}{\prod_{i=1}^{n}(s - P_i)} = -1$$

If for some reason the number of zeros of G(s)H(s) is more than the number of its poles, i.e. m > n.

m root loci in the system – m branches;

n root loci (branches) start at open-loop poles (finite poles);

$(m-n)$ root loci (branches) start at $(m-n)$ infinite poles;

all m branches end at open-loop zeros.

3 Symmetry of the RL

The RL are symmetric with respect to the real axis of the s-plane.

The roots of characteristic equation are either real or complex-conjugate.

In other words, complex roots of polynomials (i.e., characteristic equations) with real coefficients always occur as complex-conjugate pairs.

Therefore, we only need to draw the RL on the upper half s-plane and on the real-axis, the rest can be obtained by plotting its mirror image.

4 RL on the Real Axis

On a given section of the real axis, RL for K > 0 are found in the section only if the total number of poles and zeros of G(s)H(s) to the right of the section is odd.

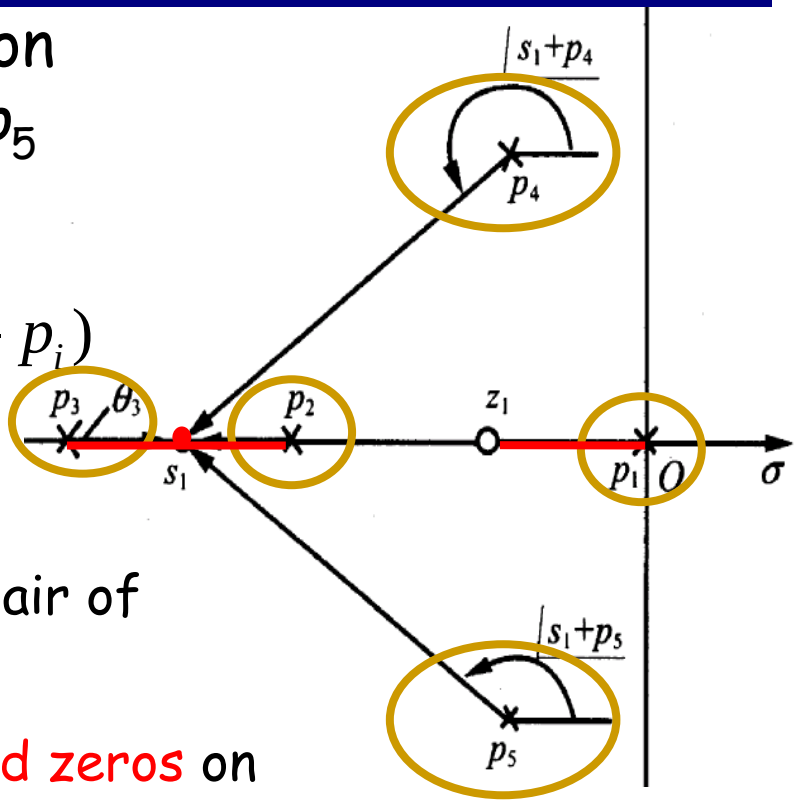
Consider the pole-zero configuration

zero: z_1 poles: p_1, p_2, p_3, p_4, p_5

Pick a test point s_1 on $[p_2, p_3]$

$$\angle G(s_1)H(s_1) = \sum_{i=1}^{1} \angle(s_1 - z_i) - \sum_{i=1}^{5} \angle(s_1 - p_i)$$

? ⇔ $\angle G(s_1)H(s_1) = (2l+1)180°$



The sum of angles provided by every pair of complex conjugate poles are 360° ;

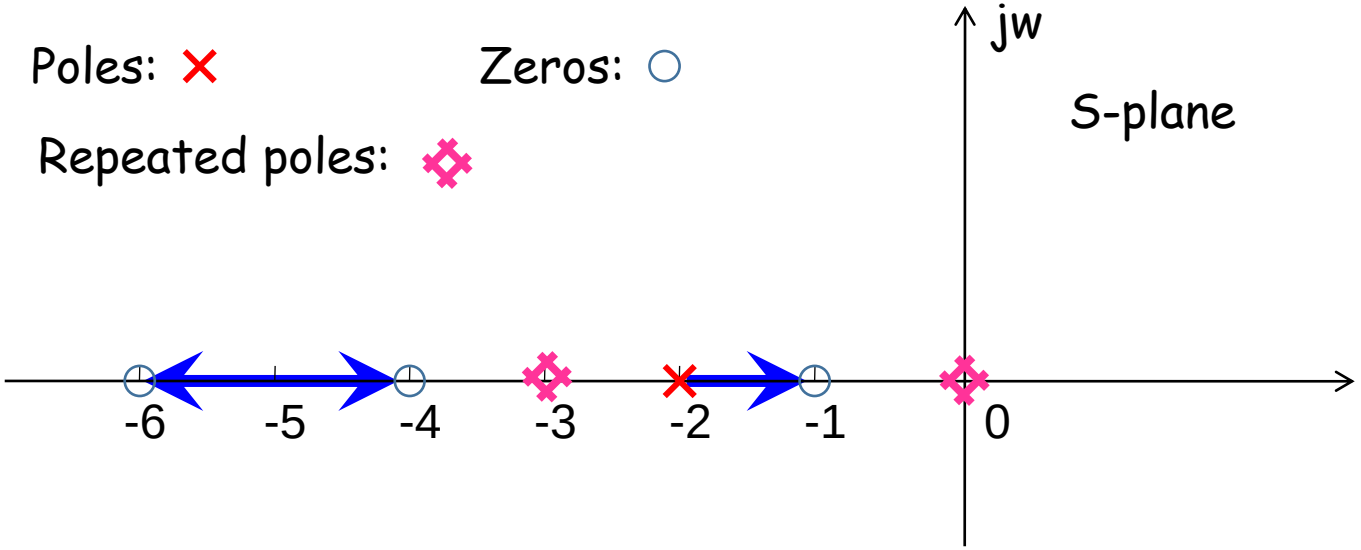
The angle provided by all the poles and zeros on the right of s_1 is 180° ;

The angle provided by all the poles and zeros on the left of s_1 is 0°.

Example Consider the following loop transfer function

$$G(s)H(s) = \frac{K(s+1)(s+4)(s+6)}{s^2(s+2)(s+3)^2}$$

Determine its root loci on the real axis.



On the right of [-2, -1] the number of real zeros and poles=3.

On the right of [-6, -4], the number of real zeros and poles=7.

5 Asymptotes of RL

When n ≠ m, there will be |n-m| asymptotes that describe the behavior of the RL at |s|=∞.

RL Equation:

$$G(s)H(s) = \frac{K\prod_{i=1}^{m}(s - Z_i)}{\prod_{i=1}^{n}(s - P_i)} = -1$$

If n > m, $(n-m)$ root loci (branches) will end at $(n-m)$ infinite zeros.

If m > n, $(m-n)$ root loci (branches) will start at $(m-n)$ infinite poles.

When n ≠ m, |n-m| branches will start (or end) at infinite poles (or zeros).

These branches will approach the infinite poles/zeros along |n-m| asymptotes.

5 Asymptotes of RL

When n ≠ m, there will be |n-m| asymptotes that describe the behavior of the RL at |s|=∞.

RL Equation:

$$G(s)H(s) = \frac{K\prod_{i=1}^{m}(s - Z_i)}{\prod_{i=1}^{n}(s - P_i)} = -1$$

The angles between the asymptotes and the real axis are (i = 0, 1, 2, … ,n-m-1):

$$\varphi_{i+1} = \frac{(2i+1)180^\circ}{|n-m|}$$

The asymptotes intersect the real axis at (the centroid):

$$\sigma_0 = \frac{\sum_{i=1}^{n} p_i - \sum_{i=1}^{m} z_i}{n-m}$$

Example Consider the following loop transfer function

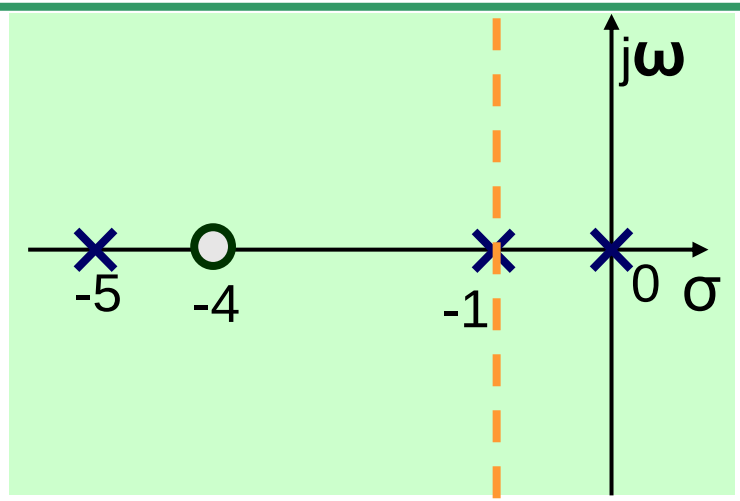
$$G(s)H(s)=\frac{K(0.25s+1)}{s(s+1)(0.2s+1)}=\frac{K^{*}(s+4)}{s(s+1)(s+5)}$$

Determine the asymptotes of its root loci.

3 poles: 0, -1, -5

1 zero: -4

n - m = 3 - 1 = 2



The asymptotes intersect the real axis at

$$\sigma_0=\frac{\sum_{i=1}^{n}p_i-\sum_{i=1}^{m}z_i}{n-m}=\frac{(0)+(-1)+(-5)-(-4)}{3-1}=-1$$

The angles between the asymptotes and the real axis are

$$\varphi_{i+1}=\frac{(2i+1)180^{\circ}}{n-m},\quad i=0,1 \qquad \varphi_1=90^{\circ},\quad \varphi_2=270^{\circ}$$

Example Consider the following loop transfer function

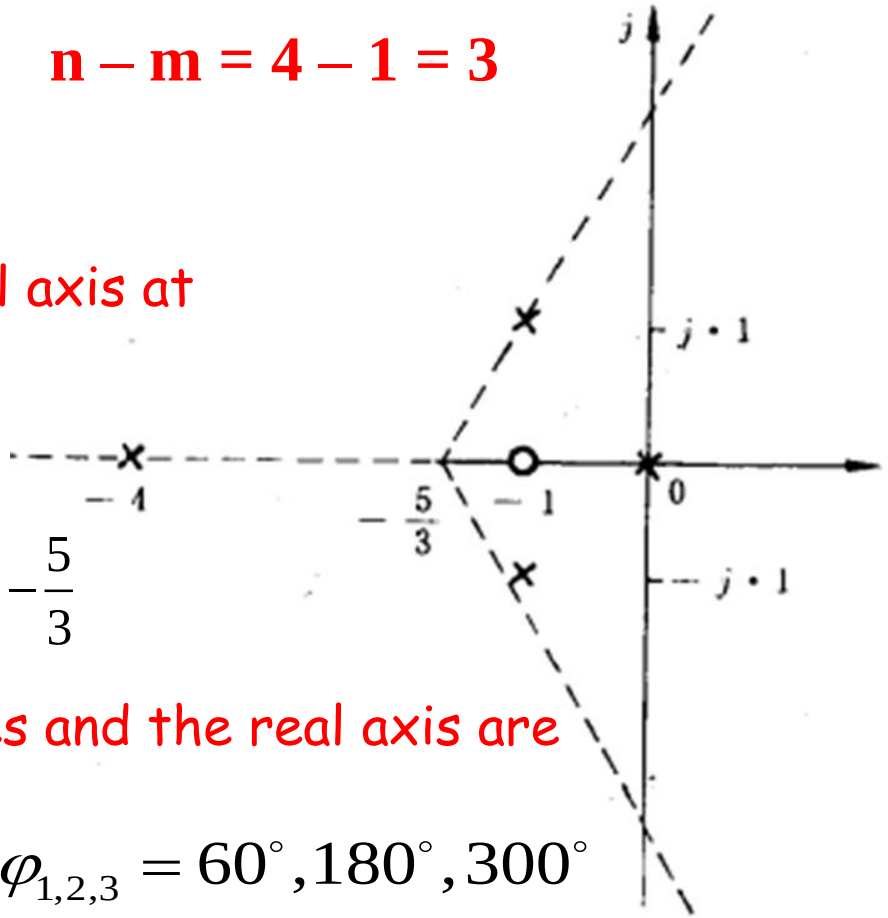
$$G(s)H(s)=\frac{K(s+1)}{s(s+4)(s^2+2s+2)}$$

Determine the asymptotes of its root loci.

4 poles: 0, -1+j, -1-j, -4

1 zero: -1

$\mathbf{n-m=4-1=3}$



The asymptotes intersect the real axis at

$$\sigma_0=\frac{\sum_{i=1}^{n}p_i-\sum_{i=1}^{m}z_i}{n-m}$$

$$=\frac{(0)+(-1+j)+(-1-j)+(-4)-(-1)}{4-1}=-\frac{5}{3}$$

The angles between the asymptotes and the real axis are

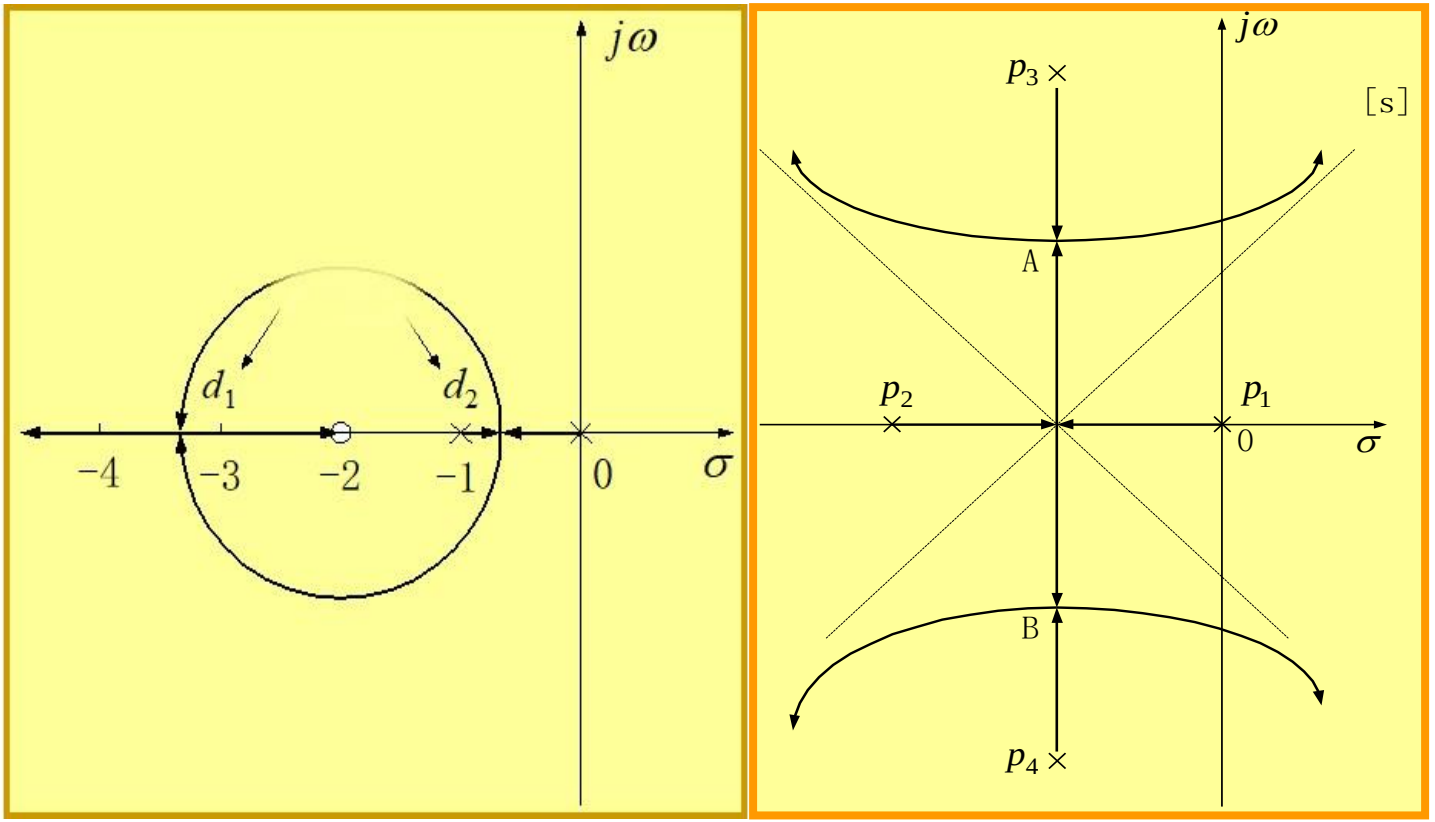
$$\varphi_{i+1}=\frac{(2i+1)180^\circ}{n-m},\quad i=0,1,2\quad \varphi_{1,2,3}=60^\circ,180^\circ,300^\circ$$

6 Breakaway or Break-in points on the RL

Breakaway and break-in points on the RL are points where two or more branches intersect.

They correspond to multi-order roots of the RL equation.

The breakaway (break-in) points on the RL are determined by finding the roots of dG(s)H(s)/ds=0.

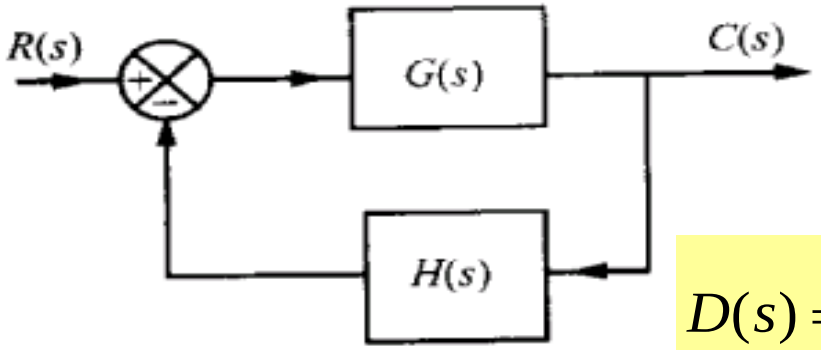


The breakaway point σ_b is the solution of

Method - 1

$$\sum_{i=1}^{m}\frac{1}{\sigma_b - z_i}=\sum_{i=1}^{n}\frac{1}{\sigma_b - p_i}$$

Proof:



$$G(s)H(s)=K\frac{\prod_{i=1}^{m}(s-z_i)}{\prod_{i=1}^{n}(s-p_i)}$$

$$D(s)=1+G(s)H(s)=\prod_{i=1}^{n}(s-p_i)+K\prod_{i=1}^{m}(s-z_i)=0$$

Suppose the multi-order root is s_1, then we have

$$D(s_1)=\prod_{i=1}^{n}(s_1-p_i)+K\prod_{i=1}^{m}(s_1-z_i)=0$$

$$\prod_{i=1}^{n}(s_1-p_i)=-K\prod_{i=1}^{m}(s_1-z_i)$$

$$\frac{d}{ds_1}D(s_1)=\frac{d}{ds_1}\left[\prod_{i=1}^{n}(s_1-p_i)+K\prod_{i=1}^{m}(s_1-z_i)\right]=0$$

$$\frac{d}{ds_1}\left[\prod_{i=1}^{n}(s_1-p_i)\right]=-K\frac{d}{ds_1}\left[\prod_{i=1}^{m}(s_1-z_i)\right]$$

$$\prod_{i=1}^{n}(s_1 - p_i) = -K\prod_{i=1}^{m}(s_1 - z_i) \quad (1)$$

$$\frac{d}{ds_1}\left[\prod_{i=1}^{n}(s_1 - p_i)\right] = -K\frac{d}{ds_1}\left[\prod_{i=1}^{m}(s_1 - z_i)\right] \quad (2)$$

dividing (2) by (1) yields

$$\frac{\dfrac{d}{ds_1}\left[\prod_{i=1}^{n}(s_1 - p_i)\right]}{\prod_{i=1}^{n}(s_1 - p_i)} = \frac{\dfrac{d}{ds_1}\left[\prod_{i=1}^{m}(s_1 - z_i)\right]}{\prod_{i=1}^{m}(s_1 - z_i)}$$

$$\frac{d}{ds_1}\ln\left[\prod_{i=1}^{n}(s_1 - p_i)\right] = \frac{d}{ds_1}\ln\left[\prod_{i=1}^{m}(s_1 - z_i)\right]$$

Thus

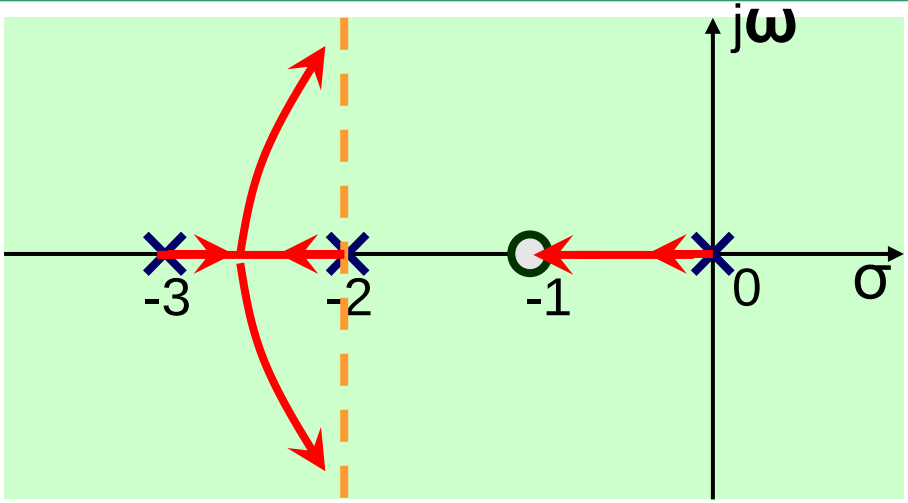
$$\sum_{i=1}^{m}\frac{1}{s_1 - z_i} = \sum_{i=1}^{n}\frac{1}{s_1 - p_i}$$

Or

$$\sum_{i=1}^{n}\frac{d}{ds_1}\ln(s_1 - p_i) = \sum_{i=1}^{m}\frac{d}{ds_1}\ln(s_1 - z_i)$$

Solution s_1 is the breakaway point σ_b

Example The poles and zeros of G(s)H(s) are shown in the following figure, determine its root loci.



Rule 1, 2, 3→

RL have three branches, starting from poles 0, -2, -3, ending at on finite zero -1 and two infinite zeros. The RL are symmetrical with respect to the real axis.

Ruel 4→ The sections [-1,0] and [-3,-2] on the real axis are on the RL.

Rule 5→ The RL have two asymptotes ($n-m=2$)

$$\varphi_{i+1}=\frac{(2i+1)180^\circ}{n-m}=90^\circ,270^\circ \qquad i=0,1$$

$$\sigma_0=\frac{\sum_{i=1}^{n}p_i-\sum_{i=1}^{m}z_i}{n-m}=\frac{0+(-2)+(-3)-(-1)}{2}=-2$$

Rule 6→ The RL have breakaway points on the real axis (within [-3,-2])

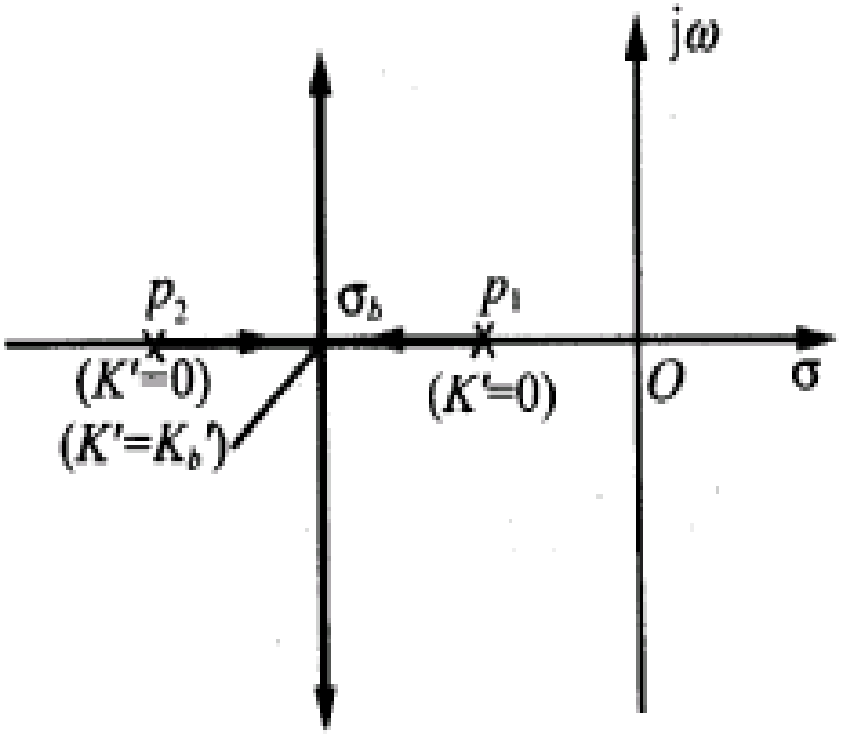
$$\frac{1}{\sigma_b+1}=\frac{1}{\sigma_b-0}+\frac{1}{\sigma_b+2}+\frac{1}{\sigma_b+3} \qquad \sigma_b=-2.47$$

The breakaway point σ_b is the solution of

Method – 2

$$\frac{d}{ds}(1+G(s)H(s)) = \frac{d}{ds}\left(\frac{\prod_{i=1}^{m}(s-z_i)}{\prod_{i=1}^{n}(s-p_i)}\right) = 0$$

Proof:



Since the break-away point is a multiple root of the equation, it must be a root of also its derivative.

→Break away points satisfy both

$$\frac{d}{ds}\left(\frac{\prod_{i=1}^{m}(s-z_i)}{\prod_{i=1}^{n}(s-p_i)}\right) = 0$$

$$1+G(s)H(s) = 0$$

The breakaway point σ_b is the solution of

Method – 2

$$\frac{d}{ds}(1+G(s)H(s)) = \frac{d}{ds}\left(\frac{\prod_{i=1}^{m}(s-z_i)}{\prod_{i=1}^{n}(s-p_i)}\right) = 0$$

Let:

$$G(s)H(s) = K\frac{\prod_{i=1}^{m}(s-z_i)}{\prod_{i=1}^{n}(s-p_i)} = K\frac{Z(s)}{P(s)},$$

$$Z(s) = \prod_{i=1}^{m}(s-z_i),$$

$$P(s) = \prod_{i=1}^{n}(s-p_i)$$

$$\frac{d}{ds}(1+G(s)H(s))$$

$$= \frac{d}{ds}\left(\frac{Z(s)}{P(s)}\right)$$

$$= \frac{Z'(s)P(s)-P'(s)Z(s)}{(P(s))^2} = 0$$

Since the break-away point is a multiple root of the equation, it must be a root of also its derivative.

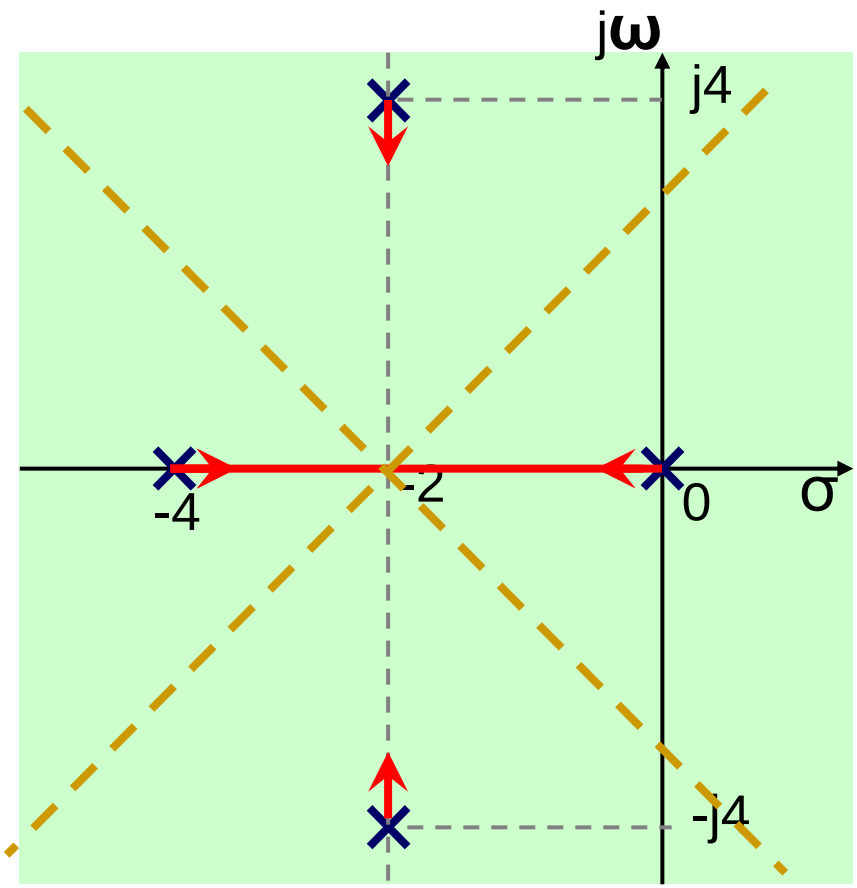
→Break away points satisfy both

$$Z'(s)P(s)-P'(s)Z(s) = 0$$

$$1+G(s)H(s) = 1+K\frac{Z(s)}{P(s)} = 0$$

Example

$$G(s)H(s)=\frac{K}{s(s+4)(s^2+4s+20)}$$



Rule 1, 2, 3, 4 → n = 4, m=0

the RL are symmetrical with respect to the real axis;
the RL has four branches which start from poles 0,-4 and $-2\pm j4$;
the RL end at infinite zeros;
the intersection [-4,0] on the real-axis is RL

Rule 5→

The RL have four asymptotes.

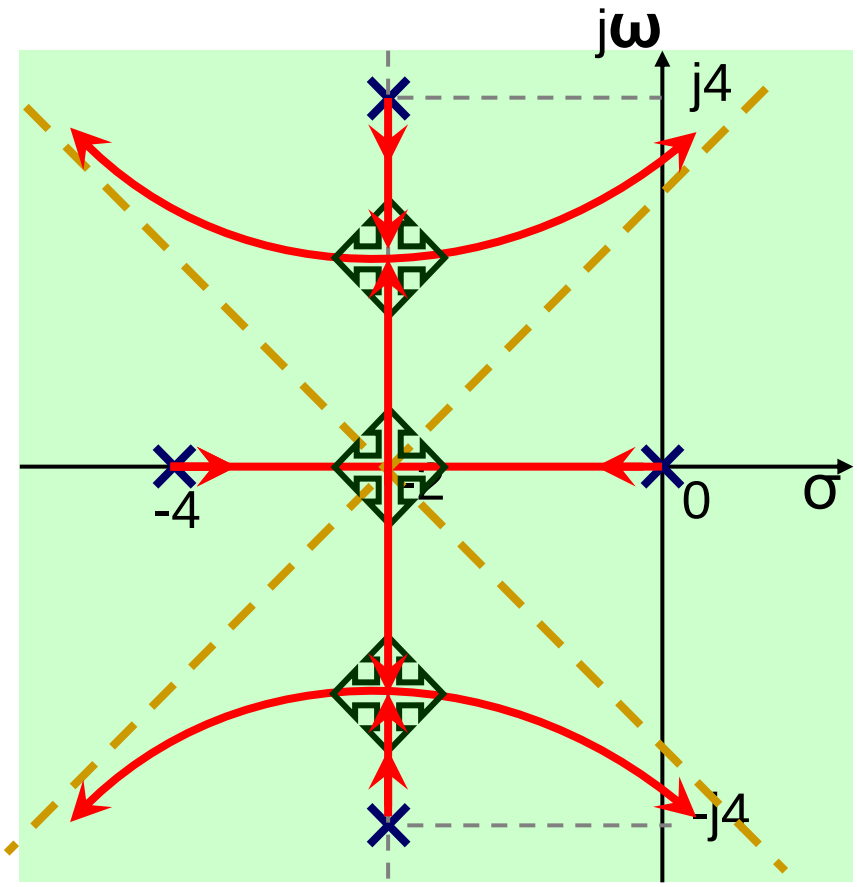
$$\varphi_{i+1}=\frac{(2i+1)180^\circ}{n-m}\quad (i=0,1,2,3)$$

$$=45^\circ,135^\circ,225^\circ,315^\circ$$

$$\sigma_0=\frac{\sum_{i=1}^{n}p_i-\sum_{i=1}^{m}z_i}{n-m}=\frac{0+(-4)+(-2)+(-2)}{4}=-2$$

Example

$$G(s)H(s)=\frac{K}{s(s+4)(s^2+4s+20)}$$



Rule 6→the breakaway point of the RL

$$1+G(s)H(s)=1+\frac{K}{s(s+4)(s^2+4s+20)}=0$$

$$1+G(s)H(s)=1+\frac{K}{s^4+8s^3+36s^2+80s}=0$$

$$\frac{d}{ds}(1+G(s)H(s))=-\frac{K\left(4s^3+24s^2+72s+80\right)}{\left(s^4+8s^3+36s^2+80s\right)^2}=0$$

$$-(4s^3+24s^2+72s+80)=0$$

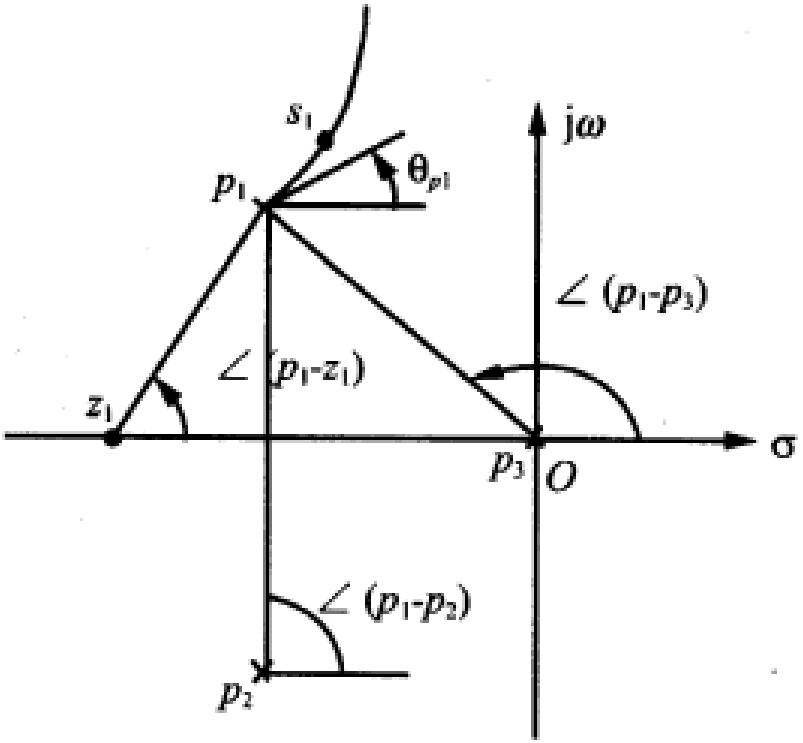
$$\sigma_{b1}=-2$$

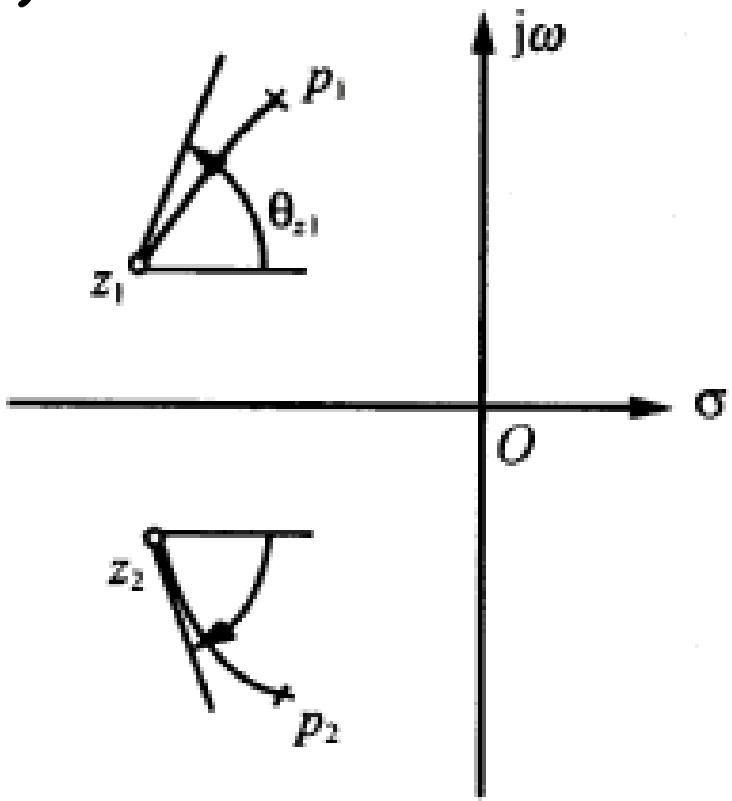
$$\sigma_{b2,\ 3}=-2\pm j2.45$$

7 Angles of departure and angles of arrival of the RL

When the locus departs from a pole or arrives at a zero, from which direction does it so?

The angle of departure or arrival of a root locus at a pole or zero of G(s)H(s) respectively, denotes the angle of the tangent to the locus near the point (i.e., pole or zero).





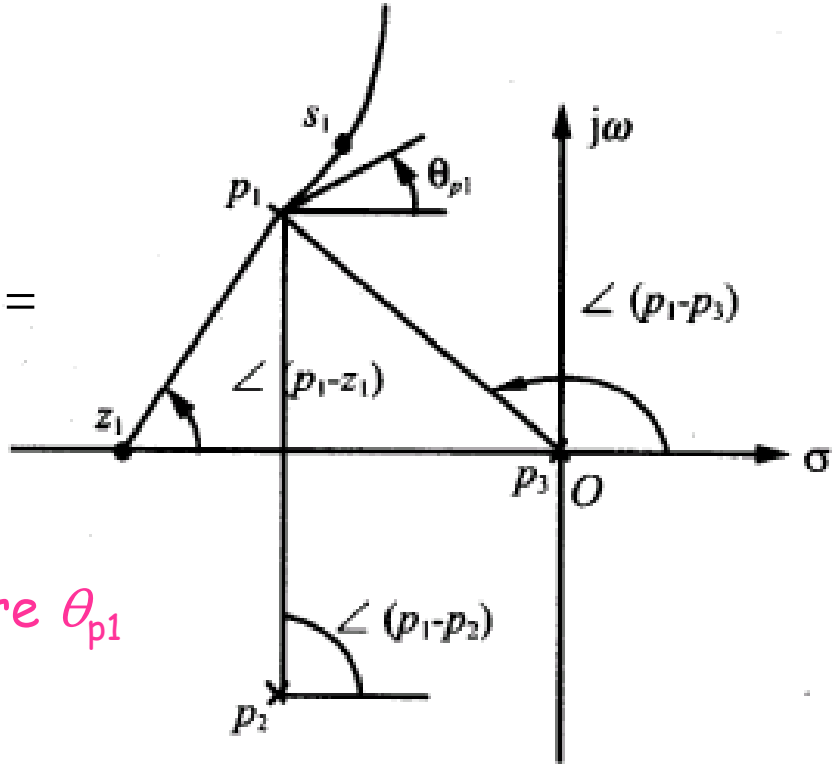
Angle of Departure:

$$\theta_{pj} = (2l+1)\pi + \sum_{i=1}^{m} \angle(p_j - z_i) - \sum_{\substack{i=1 \\ i \neq j}}^{n} \angle(p_j - p_i), \quad l = 0, \pm 1, \cdots$$

Pick up a point s_1 that is close to p_1

Applying Angle Equation (AE)

$$\angle(s_1 - z_1) - \angle(s_1 - p_1) - \angle(s_1 - p_2) - \angle(s_1 - p_3) = (2l+1)\pi$$

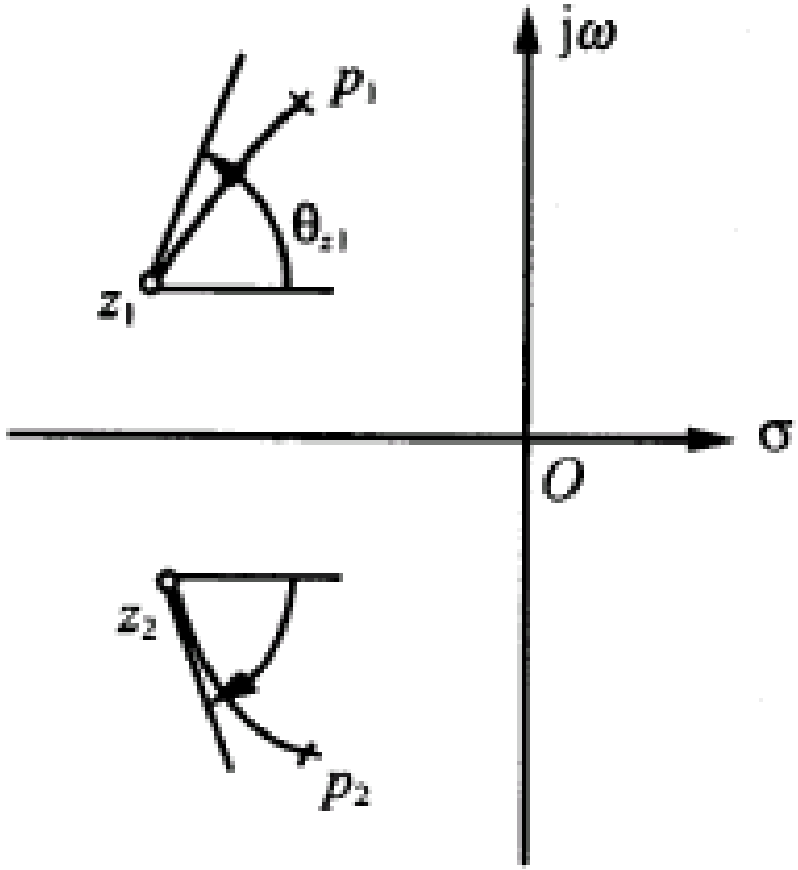


$s_1 \rightarrow p_1$ $\angle(s_1 - p_1)$ →angle of departure θ_{p1}

$$\theta_{p1} = (2l+1)\pi + \angle(p_1 - z_1) - \angle(p_1 - p_2) - \angle(p_1 - p_3)$$

Angle of Arrival:

$$\varphi_{zj}=(2l+1)\pi+\sum_{i=1}^{n}\angle(z_j-p_i)-\sum_{\substack{i=1\\ i\neq j}}^{m}\angle(z_j-z_i),\quad l=0,\pm1,\cdots$$



Example Consider the following loop transfer function

$$G(s)H(s)=\frac{Ks}{\left(s^2+2s+2\right)\left(s-1\right)}$$

Determine its RL when K varies from 0 to ∞.

3 poles $P_{1,2}$ = -1±j, P_3=1; 1 zero Z_1=0, n-m=2.
3 branches , 2 asymptotes

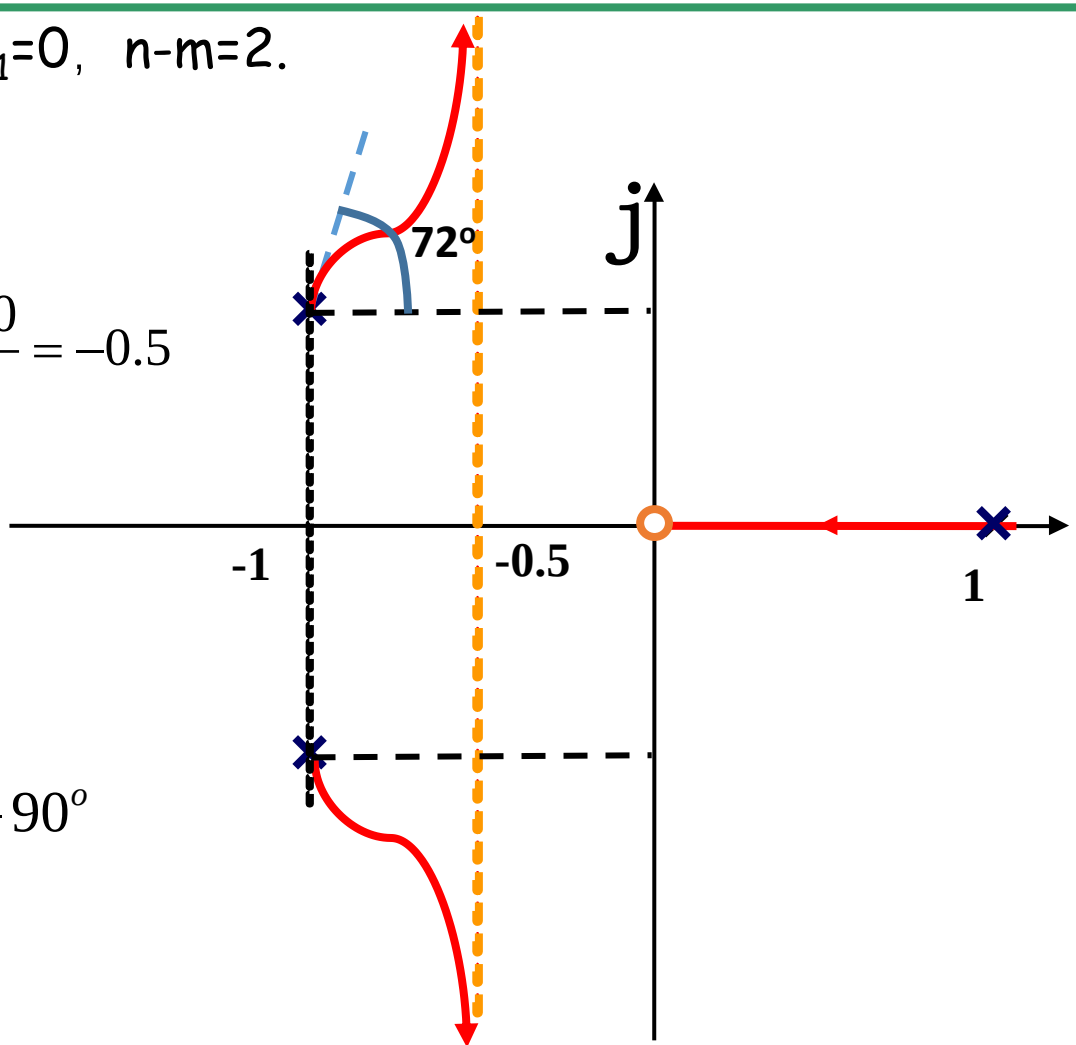
$$\sigma_0=\frac{\sum_{i=1}^{n}P_i-\sum_{i=1}^{m}Z_i}{n-m}=\frac{-1+j-1-j+1-0}{3-1}=-0.5$$

$$\varphi_{i+1}=\frac{(2i+1)\pi}{n-m}=\frac{\pi}{2},\frac{3\pi}{2}$$

Angle of departure:

$$\theta_{pl}=(2l+1)\times180^o+135^o-153^o-90^o$$

$$=72^o$$



8 Intersection of the RL with the Imaginary Axis

Intersection of the RL with Im-axis→?

The characteristic equation have roots on the Im-axis and the system is marginally stable.

Method 1 Use Routh's criterion to obtain the value of K when the system is marginally stable, then get w from K.

Method 2

$$s = j\omega \rightarrow 1 + G(s)H(s) = 0 \rightarrow 1 + G(j\omega)H(j\omega) = 0$$

$$\begin{cases} \mathrm{Re}\left[1 + G(j\omega)H(j\omega)\right] = 0 \\ \mathrm{Im}\left[1 + G(j\omega)H(j\omega)\right] = 0 \end{cases} \rightarrow \omega, K$$

Example Consider the following loop transfer function

$$G(s)H(s) = \frac{K}{s(s+1)(s+2)}$$

Determine the intersection of the RL with Im-axis.

Method 1

Closed-loop CE: $s(s+1)(s+2) + K = s^3 + 3s^2 + 2s + K = 0$

Routh's Tabulation:

s^3	1	2
s^2	3	K
s^1	$\frac{6-K}{3}$	0
s^0	K	

Marginally stable:
K =6

Auxiliary equation:

$$3s^2 + 6 = 0$$

Intersection point: $s = \pm j\sqrt{2}$

Example Consider the following loop transfer function

$$G(s)H(s)=\frac{K}{s(s+1)(s+2)}$$

Determine the intersection of the RL with Im-axis.

Method 2

Closed-loop CE: $s(s+1)(s+2)+K=s^3+3s^2+2s+K=0$

$s=j\omega$ →Closed-loop CE $\rightarrow 1+G(j\omega)H(j\omega)=0$

$$(j\omega)^3+3(j\omega)^2+2(j\omega)+K=(K-3\omega^2)+j(2\omega-\omega^3)=0$$

Real part: $K-3\omega^2=0$

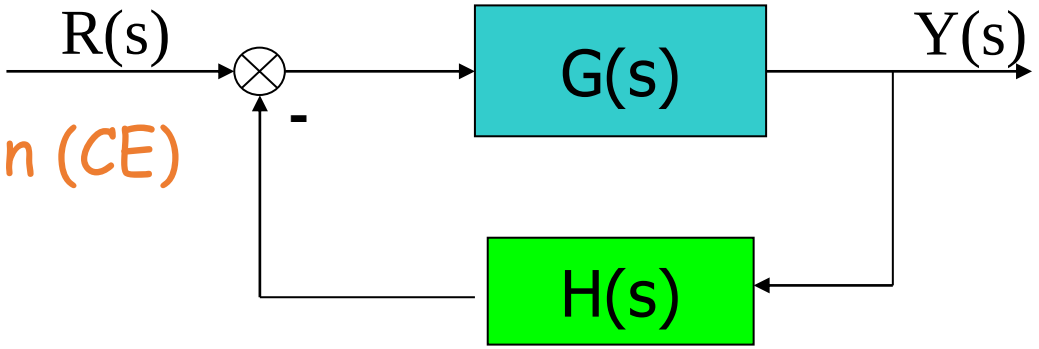
Imaginary part: $2\omega-\omega^3=0$

$\Rightarrow \quad \omega=\pm\sqrt{2} \quad K=6$

Summary of General Rules for Constructing Root Loci

First, obtain the characteristic equation (CE)

$$1+G(s)H(s)=0$$



Then, rearrange the equation so that the parameters of interest appear as the multiplying factor in the form

$$1+\frac{K(s-z_1)(s-z_2)\cdots(s-z_m)}{(s-p_1)(s-p_2)\cdots(s-p_n)}=0$$

We sketch the Root Loci when K > 0 varies from 0 to ∞.

Step 1: Locate the poles and zeros of G(s)H(s) in the s-plane. The RL start from poles of G(s)H(s) and terminate at zeros (finite zeros or zeros at infinity)

Step 2: Determine the RL on the real-axis. Choose a test point on the real-axis, if the total number of real poles and real zeros to the right of the test is odd, then the test point is on the RL.

Step 3: Determine the asymptotes of RL.

The angles between the asymptotes and the real axis are (i= 0, 1, 2, … ,n-m-1) :

$$\varphi_{i+1} = \frac{(2i+1)180^\circ}{n-m}$$

The asymptotes intersect the real-axis at the centroid:

$$\sigma_0 = \frac{\sum_{i=1}^{n} p_i - \sum_{i=1}^{m} z_i}{n-m}$$

Step 4: Determine the breakaway and break-in points.

The breakaway and break-in points can be determined by finding the roots of

$$\frac{d}{ds}G(s)H(s)=0$$

$$\sum_{i=1}^{m}\frac{1}{s_1-z_i}=\sum_{i=1}^{n}\frac{1}{s_1-p_i}$$

Step 5: Determine the angle of departure (angle of arrival) of the RL from a complex pole (at a complex zero)

Angle of departure from a complex pole p_j=

$$(2l+1)\times 180^o-\sum_{i=1,i\neq j}^{n}\angle(p_j-p_i)+\sum_{i=1}^{m}\angle(p_j-z_i)$$

Angle of arrival at a complex zero z_k=

$$(2l+1)\times 180^o+\sum_{i=1}^{n}\angle(z_k-p_i)-\sum_{i=1,i\neq k}^{m}\angle(z_k-z_i)$$

Step 6: Find the points where the RL may cross the imaginary-axis.

Method 1 Use Routh's criterion to obtain the value of K when the system is marginally stable, the get w from K.

Method 2

$$s = j\omega \rightarrow 1 + G(s)H(s) = 0 \rightarrow 1 + G(j\omega)H(j\omega) = 0$$

$$\begin{cases} \mathrm{Re}\left[1 + G(j\omega)H(j\omega)\right] = 0 \\ \mathrm{Im}\left[1 + G(j\omega)H(j\omega)\right] = 0 \end{cases} \rightarrow \omega, K$$

Example

Consider a unity feedback system with the open-loop transfer function as:

$$G(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Sketch RL of the closed-loop system when K varies from 0 to ∞.

Solution.

loop TF: $$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Step 1: Locate the poles and zeros of G(s)H(s) on the s-plane.

loop poles: $p_1=0$, $p_2=-3$, $p_{3,4}=-1\pm j$, $n=4$;

no loop zeros: $m=0$;

$$G(s)H(s)=\frac{K}{s(s+3)(s^2+2s+2)}$$

Step 2: Determine the RL on the real-axis. The section [0,-3] of the real axis are RL.

Step 3: Determine the asymptotes of RL

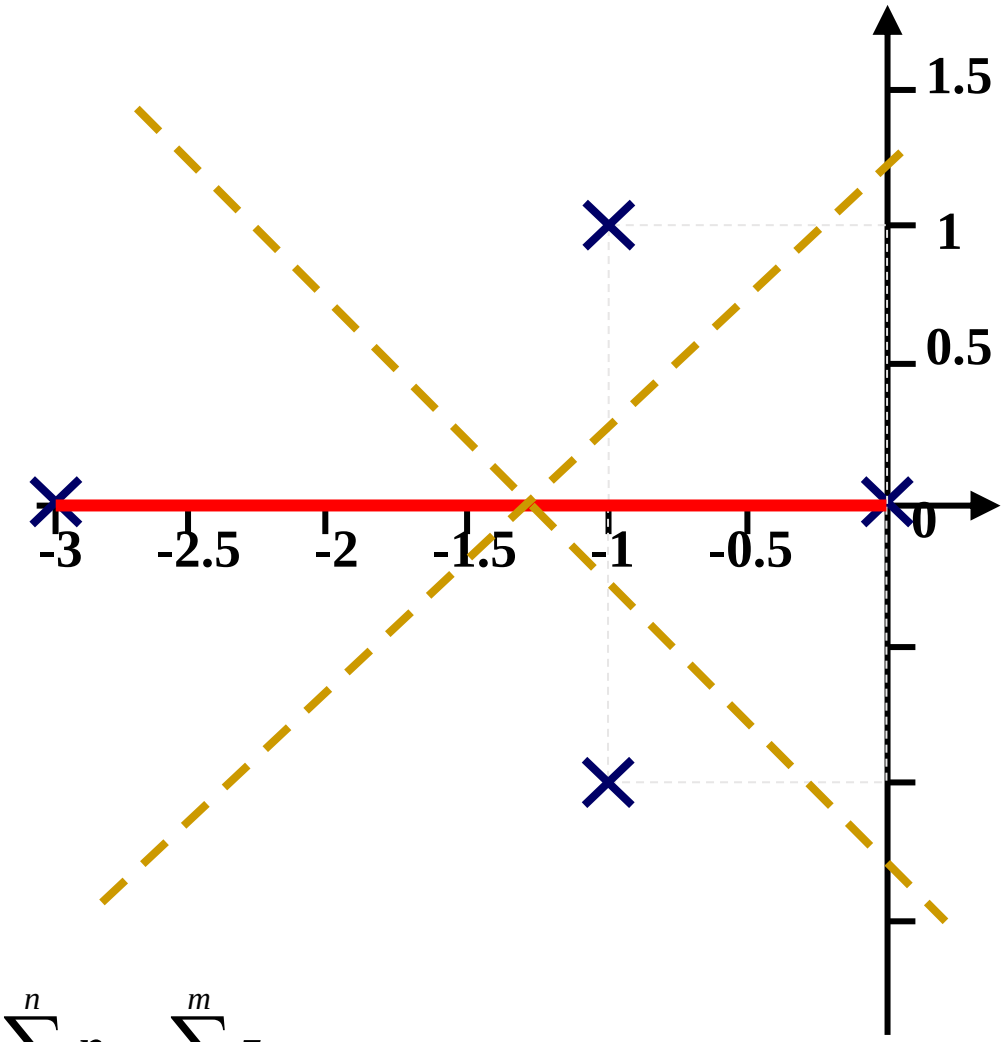
The angles between the asymptotes and the real axis are (i= 0, 1, 2, … ,n-m-1) :

$$\varphi_{i+1}=\frac{(2i+1)180^\circ}{n-m}$$

$$=45^\circ,135^\circ,225^\circ,315^\circ$$

The asymptotes intersect the real-axis at:

$$\sigma_0=\frac{\sum_{i=1}^{n}p_i-\sum_{i=1}^{m}z_i}{n-m}=-1.25$$



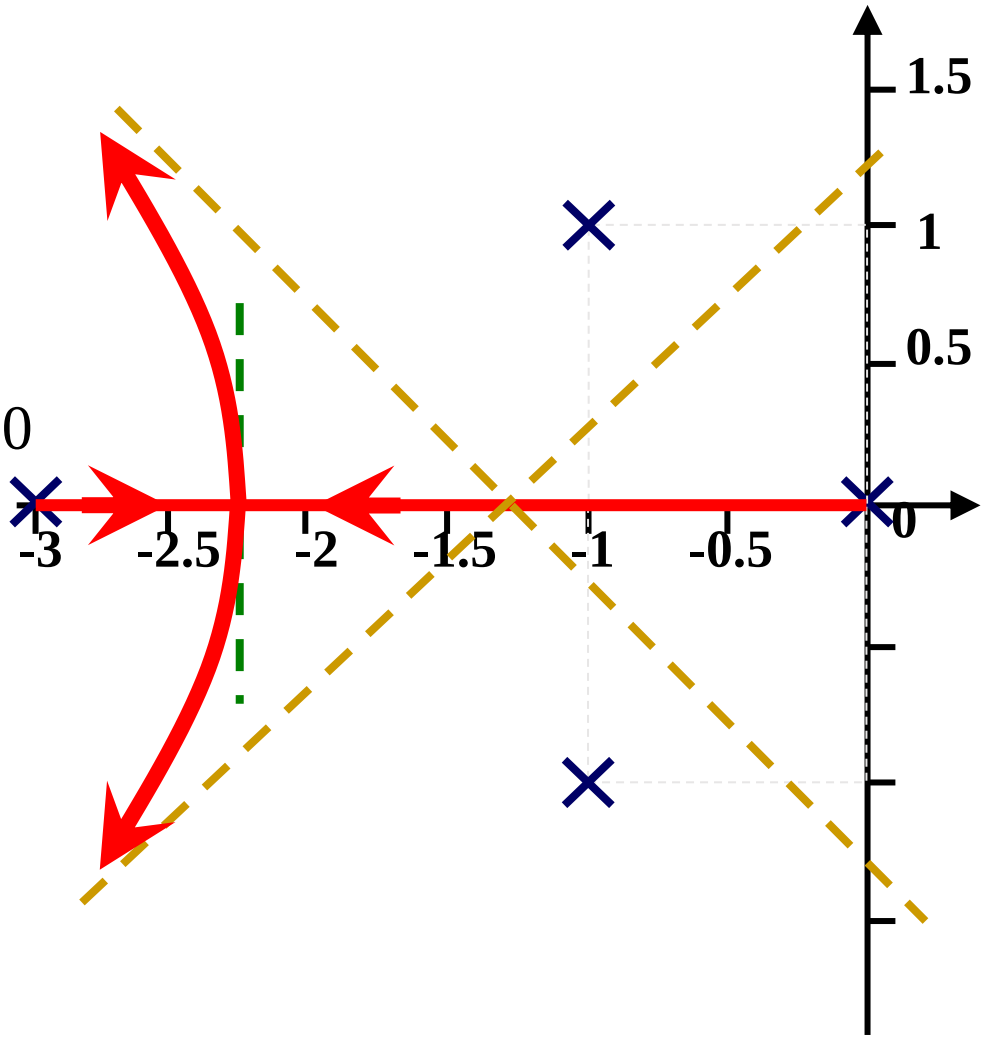
Step 4: Determine the breakaway and break-in points.

$$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

$$\sum_{i=1}^{n} \frac{1}{d - P_i} = 0, \text{ i.e.,}$$

$$\frac{1}{d} + \frac{1}{d+3} + \frac{1}{d+1-j} + \frac{1}{d+1+j} = 0$$

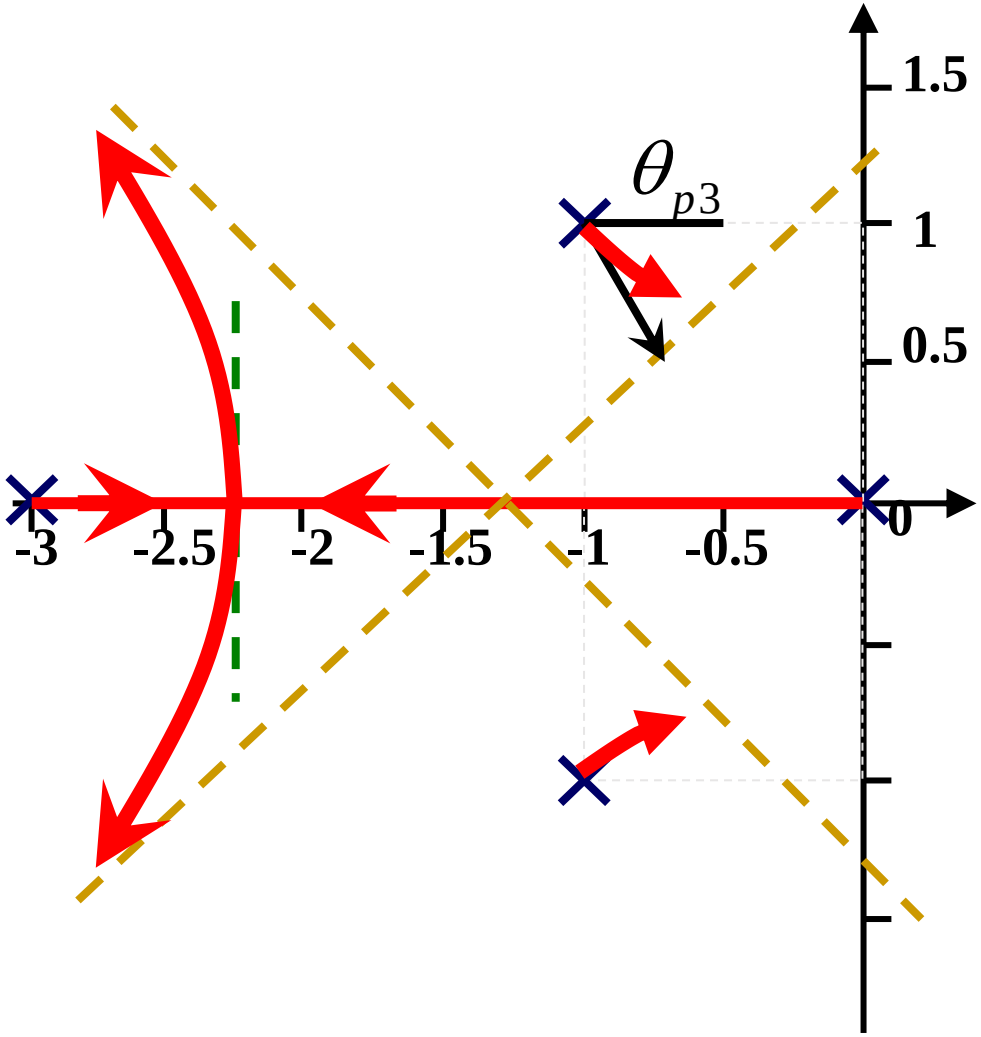
$d = -2.3$



Step 5: Determine the angle of departure (angle of arrival) of the RL from a complex pole (at a complex zero)

$$G(s)H(s)=\frac{K}{s(s+3)(s^2+2s+2)}$$

$$\theta_{p3}=180^\circ+\sum_{i=1}^{m}\angle(p_3-z_i)-\sum_{\substack{i=1\\ i\neq 3}}^{n}\angle(p_3-p_i)$$



Using measurement to estimate

$$\theta_{p3}=-71.6^\circ$$

Step 6: Find the points where the RL may cross the imaginary-axis.

$$G(s)H(s)=\frac{K}{s(s+3)(s^2+2s+2)}$$

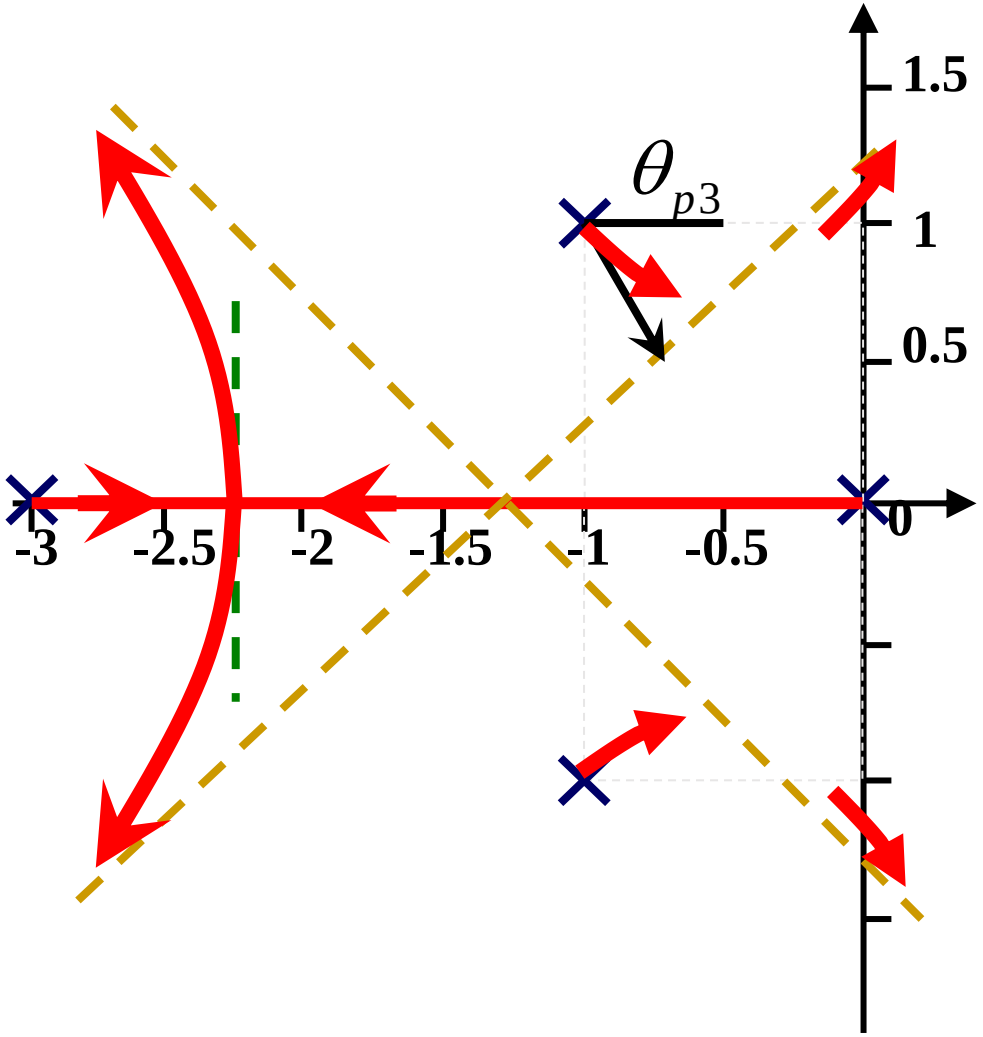
The characteristic equation:

$$D(s)=s^4+5s^3+8s^2+6s+K=0$$

Substituting s=jw into the above equation yields:

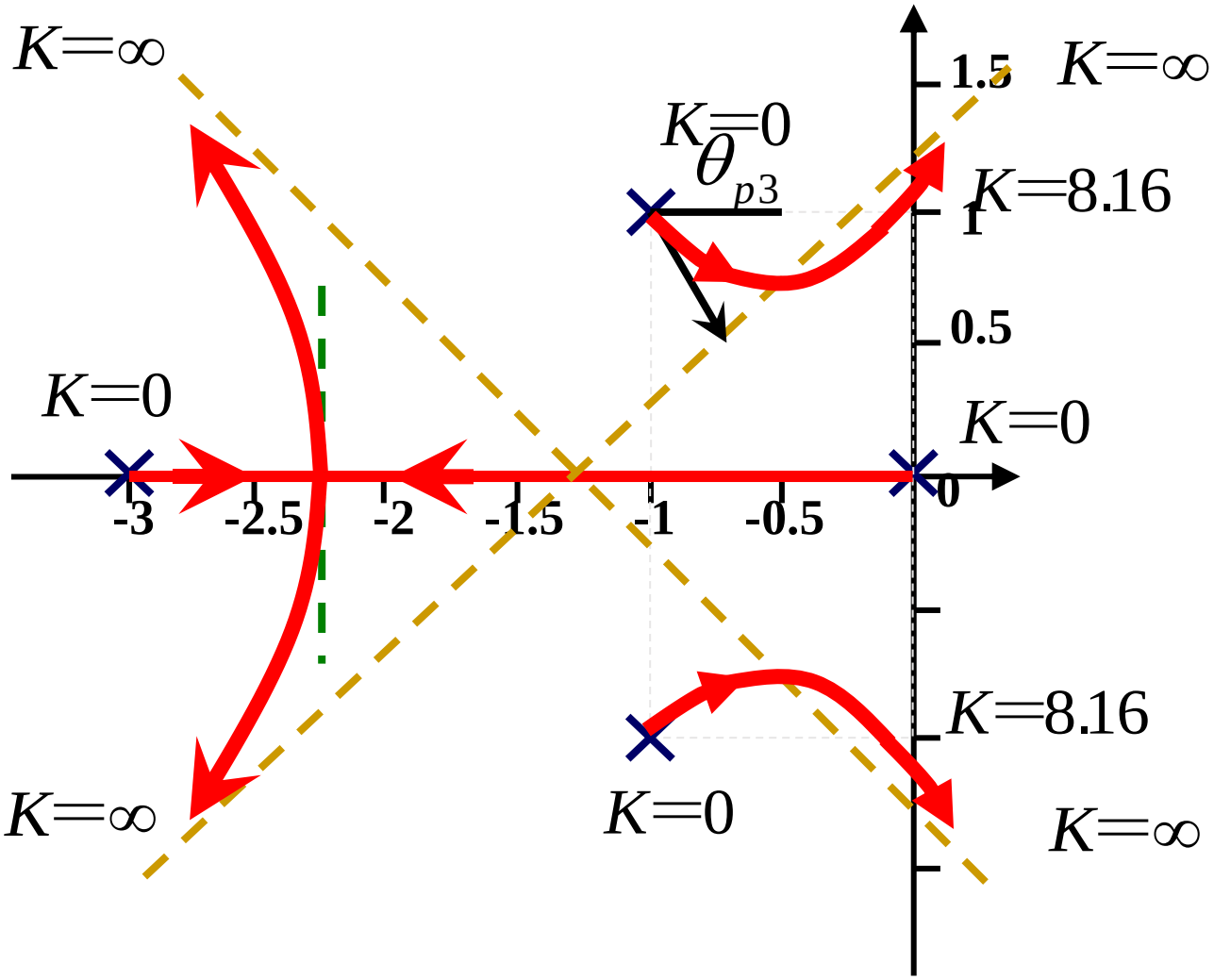
$$\begin{cases}\omega^4-8\omega^2+K=0\\ -5\omega^3+6\omega=0\end{cases}$$

$$\Rightarrow\begin{cases}\omega=\pm 1.1\\ K=8.16\end{cases}$$

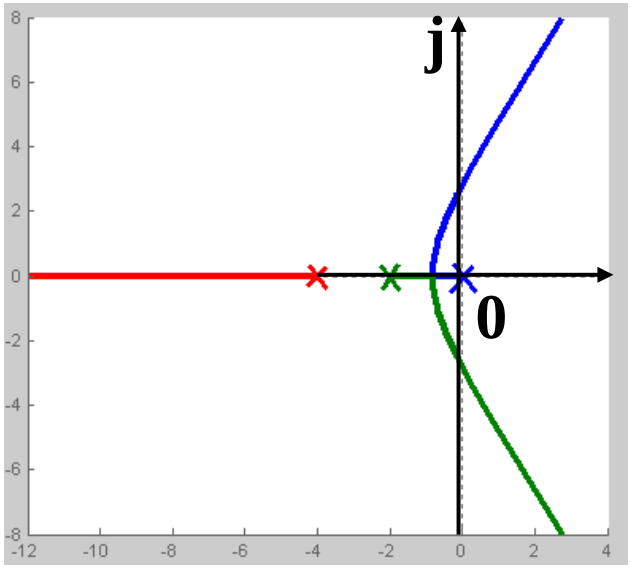


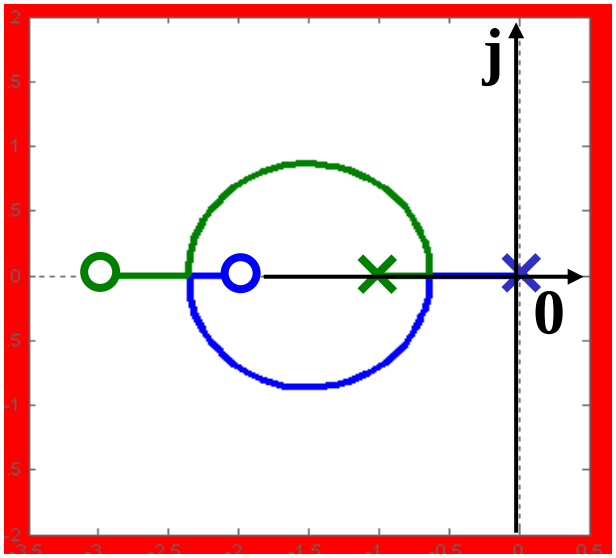
The complete RL:

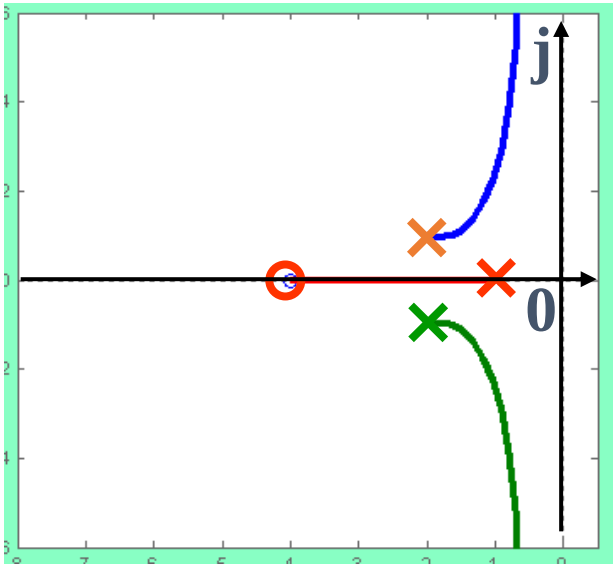
$$G(s)H(s)=\frac{K}{s(s+3)(s^2+2s+2)}$$

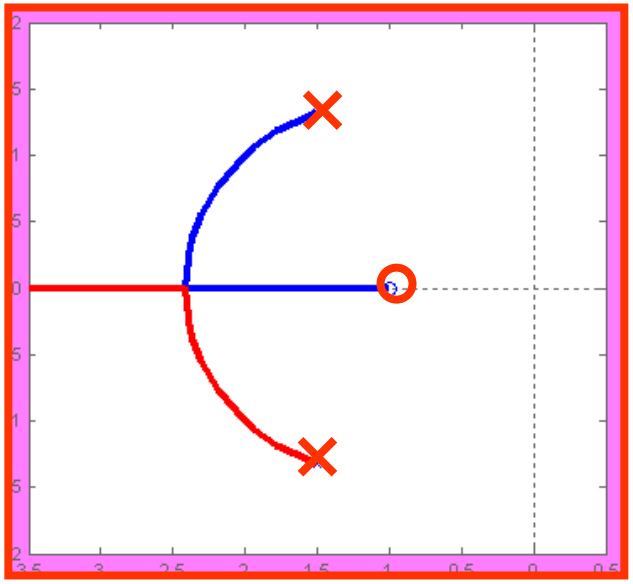


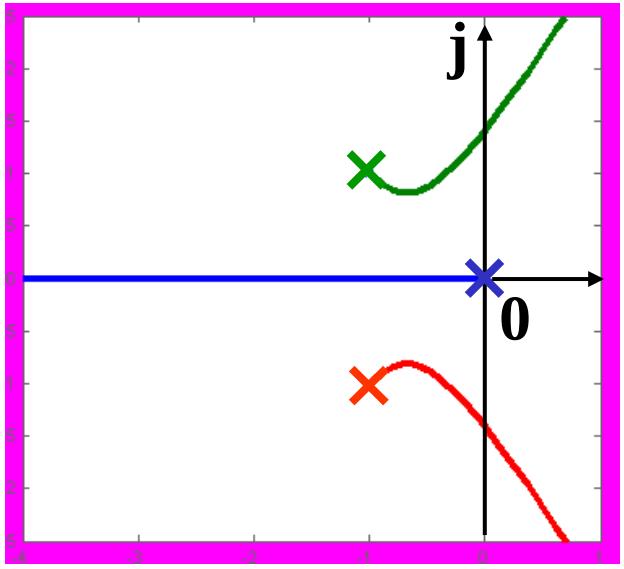
RL Demo 1

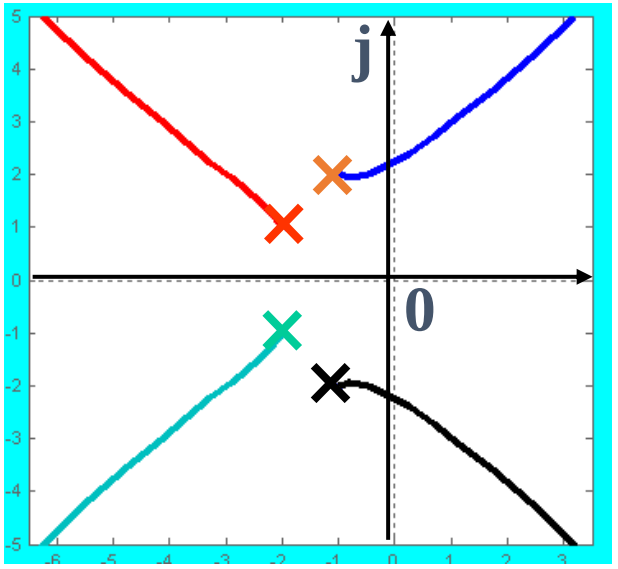




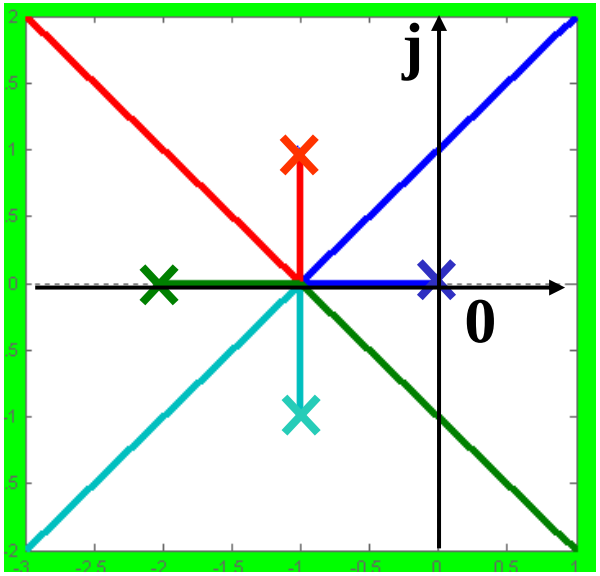


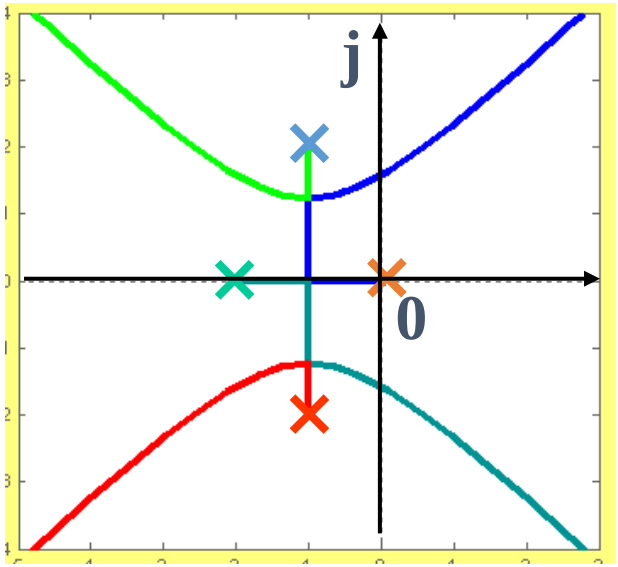


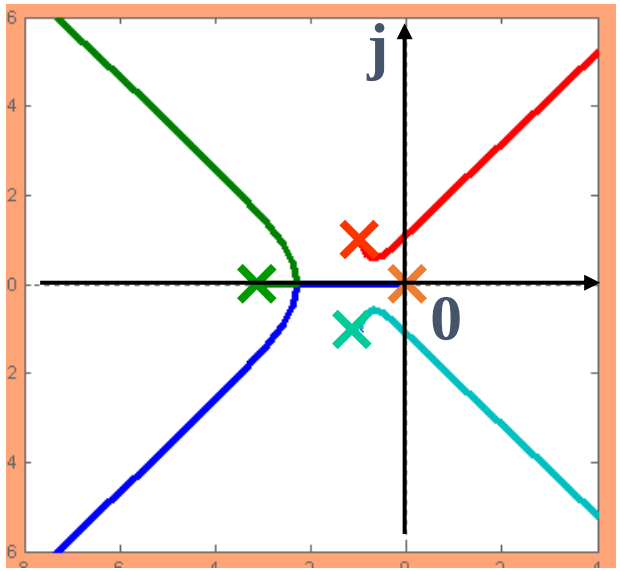


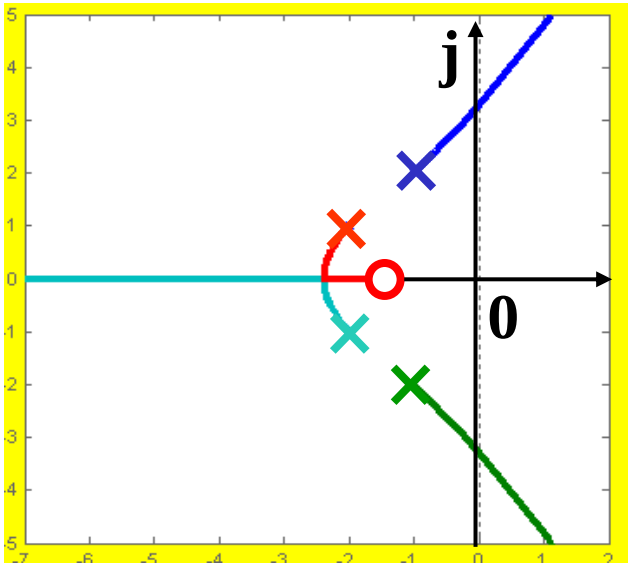


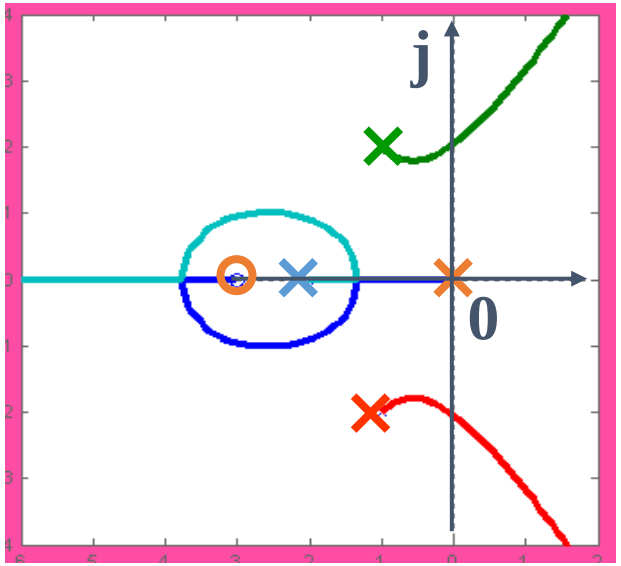
RL Demo 2

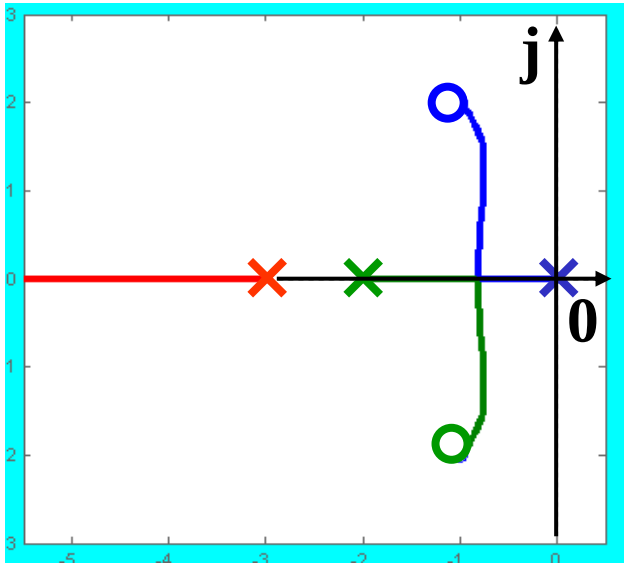












Draw RL using MATLAB

Loop function:

$$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)} = K \cdot \frac{1}{s^4+5s^3+8s^2+6s}$$

```
>>n=1;

>> d=[1 5 8 6 0];

>>sys=tf(n,d);

>>rlocus(sys)
```

Determination of closed-loop poles

Every point on the RL represents a closed-loop pole corresponding to a certain value of K which can be determined according to the Magnitude Equation.

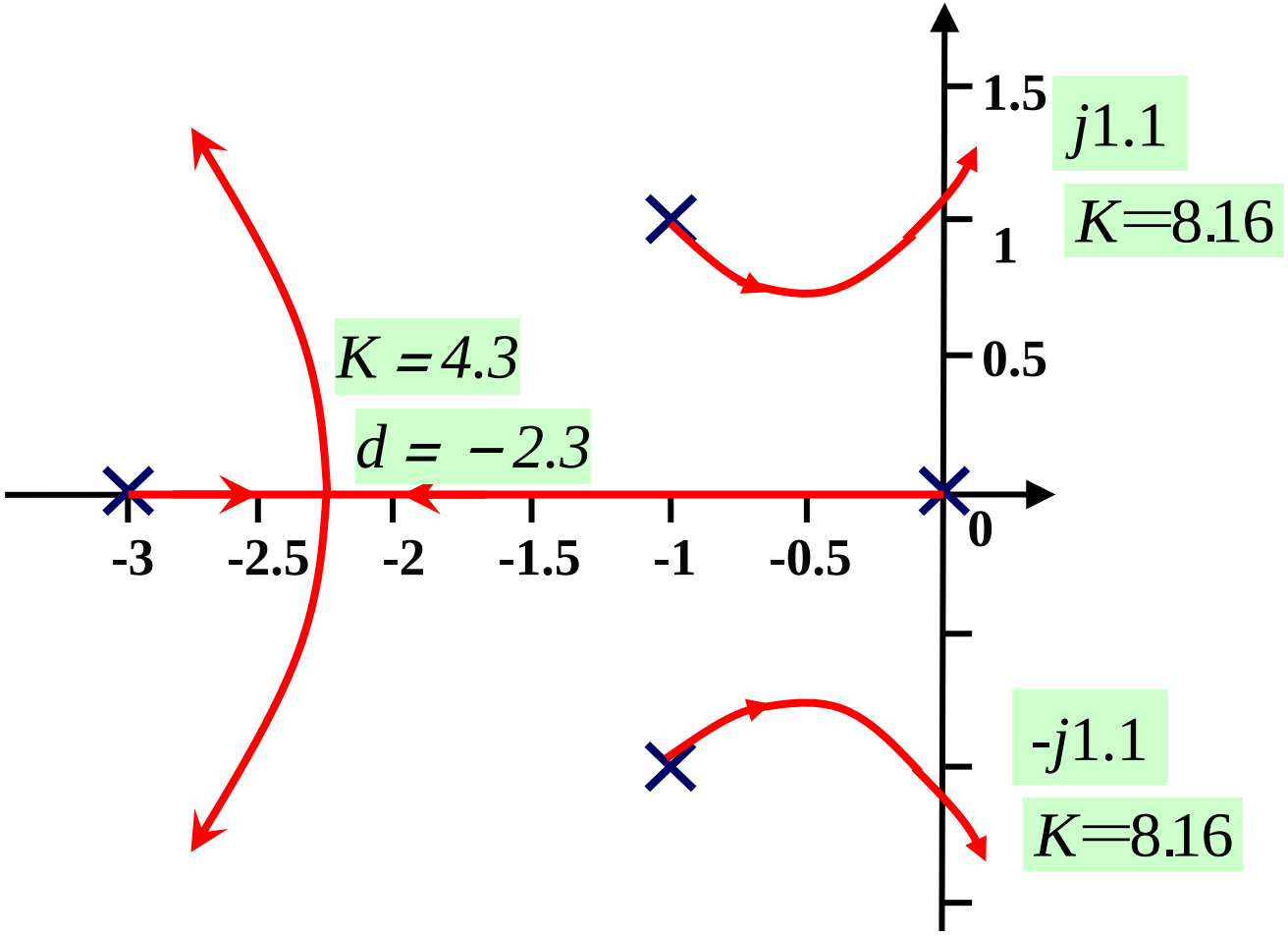
Magnitude Equation :

$$\frac{K\prod_{i=1}^{m}|s-Z_i|}{\prod_{i=1}^{n}|s-P_i|}=1$$

Example Given the RL of a unity feedback system whose open-loop transfer function is:

$$G(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Determine the closed-loop poles when K =4.



Determine the closed-loop pole according to ME:

$$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

4 poles, no zero

Magnitude Equation :

$$\frac{K\prod_{i=1}^{m}|s-Z_i|}{\prod_{i=1}^{n}|s-P_i|}=1$$

At the breakaway point d=-2.3, and its corresponding K=4.3
K increases from 0 to 4.3 before break away

So, there are the two real closed-loop poles locating at two sides of the breakaway point corresponding to K=4

Pick two points within [– 3, -2.3] and [-2.3,0] , using trial-and-error to find the closed-loop poles on the real-axis.

$$K=\prod_{i=1}^{4}|s-P_i|=4$$

$$\Leftrightarrow \left|s(s+3)(s^2+2s+2)\right|=4$$

$$s_1=-2, s_2=-2.52$$

Closed-loop CE: $s^4 + 5s^3 + 8s^2 + 6s + K = 0$

using $s_1 = -2, s_2 = -2.52$ and $K = 4$

$$(s+2)(s+2.52)(s-s_3)(s-s_4) = s^4 + 5s^3 + 8s^2 + 6s + 4$$

$$(s^2 + 4.52s + 5.04)(s-s_3)(s-s_4) = s^4 + 5s^3 + 8s^2 + 6s + 4$$

Using polynomial division to get

$$(s-s_3)(s-s_4) = s^2 + 0.48s + 0.79$$

$$\Rightarrow \begin{cases} s_3 = -0.24 + j0.86 \\ s_4 = -0.24 - j0.86 \end{cases}$$

So, when K=4, the closed-loop transfer function is

$$T(s) = \frac{4}{(s+2)(s+2.52)(s+0.24+j0.86)(s+0.24-j0.86)}$$

Effects of Adding Poles and Zeros to G(s)H(s)

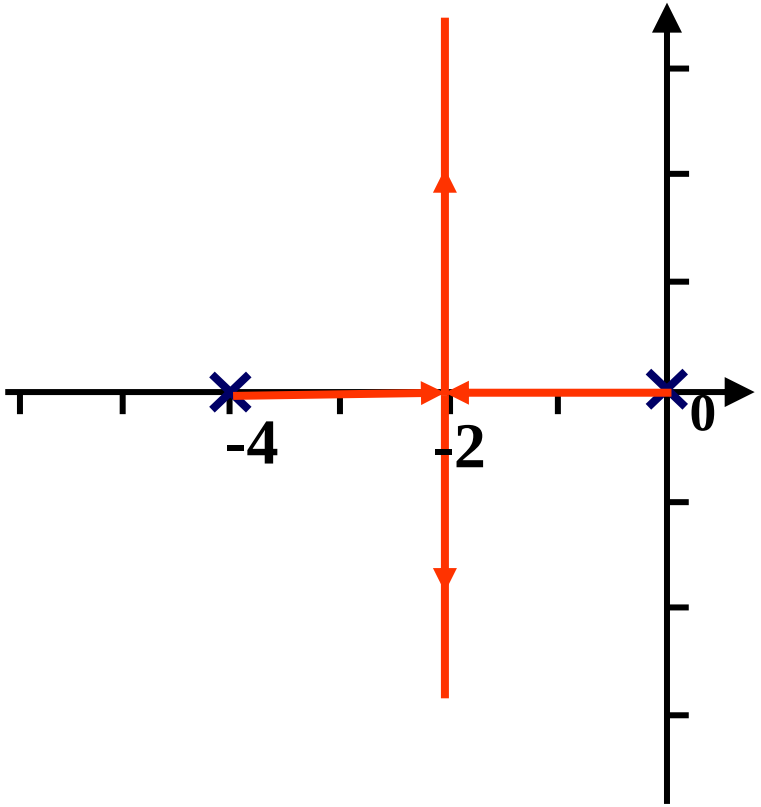
Example

$$G(s)H(s) = \frac{K}{s(s+4)}$$

Poles: -4, 0

Breakaway point:

$$\frac{dG(s)H(s)}{ds} = 0 \quad \Rightarrow \quad s = -2$$

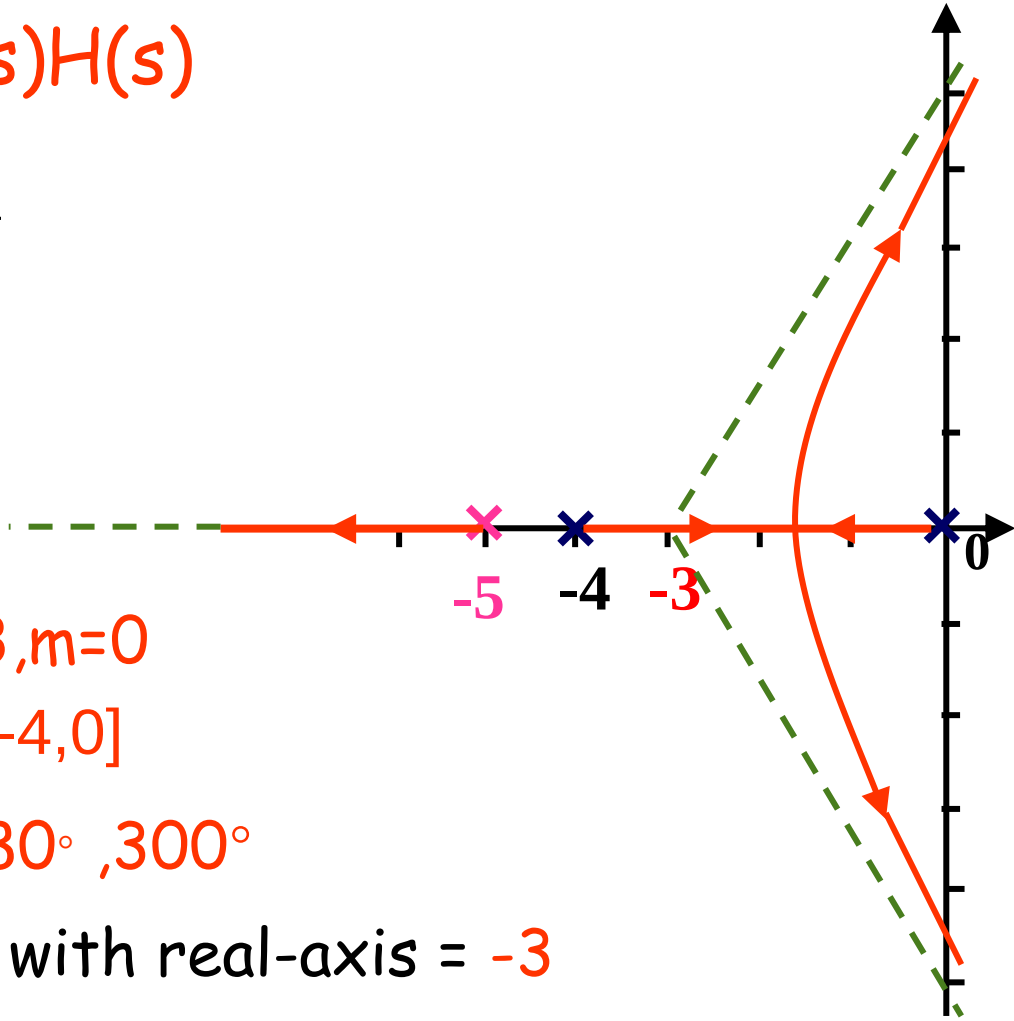


1. Addition of poles to G(s)H(s)

Example $G(s)H(s)=\dfrac{K}{s(s+4)}$

Add a pole at s=-5, we have

$$G(s)H(s)=\frac{K}{s(s+4)(s+5)}$$



Poles: -5,-4, 0 no zeros, n=3,m=0

RL on real-axis: (-∞,-5] and [-4,0]

Angle of asymptotes= 60°,180° ,300°

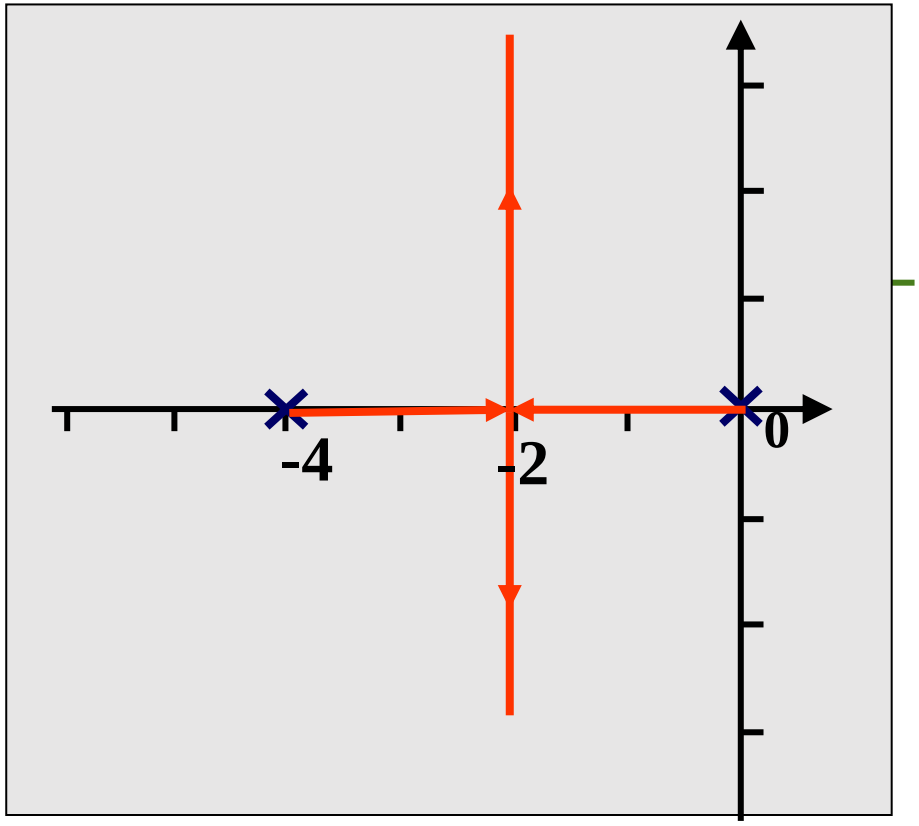
Intersection of asymptotes with real-axis = -3

Breakaway point: $\dfrac{dG(s)H(s)}{ds}=0 \quad \Rightarrow \quad s=-1.47$

Intersection of RL with Im-axis = ±4.47j

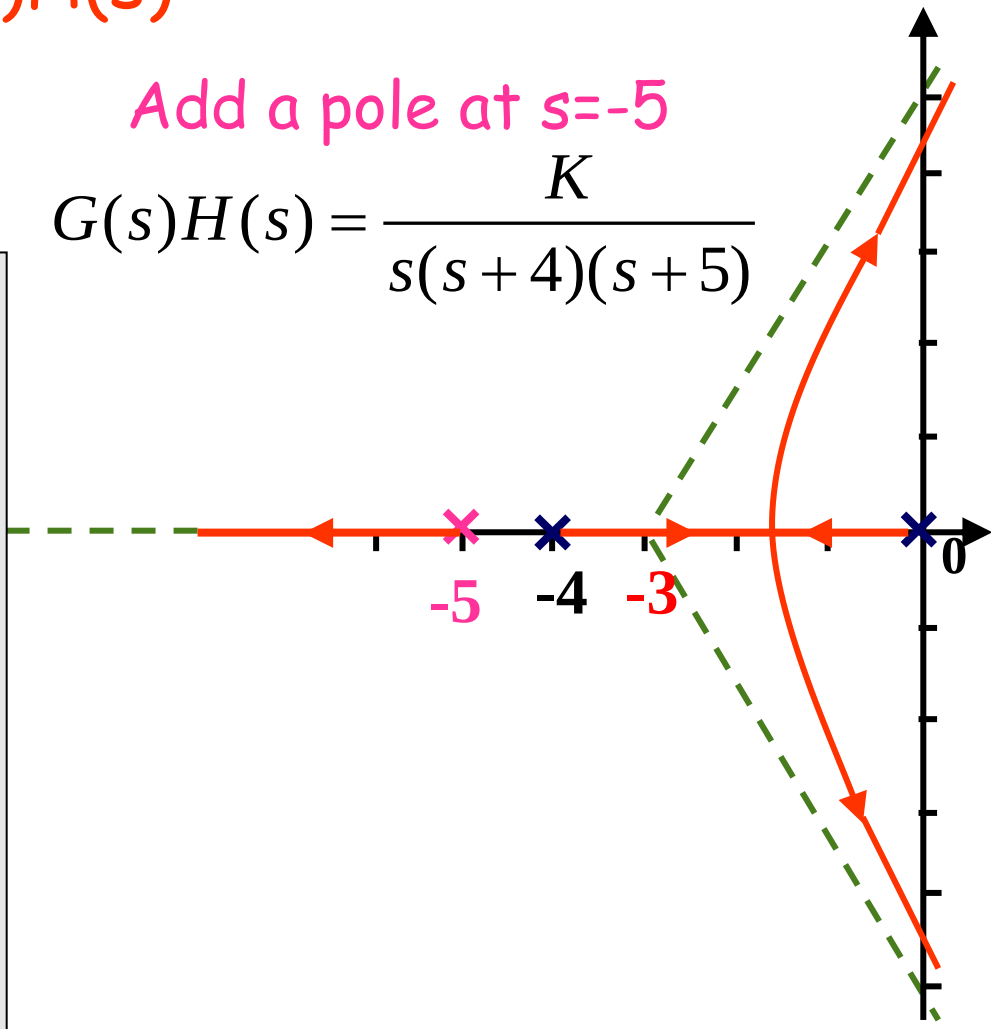
1. Addition of poles of G(s)H(s)

Example $G(s)H(s) = \dfrac{K}{s(s+4)}$



Add a pole at s=-5

$$G(s)H(s) = \frac{K}{s(s+4)(s+5)}$$



Adding a pole to G(s)H(s) has the effect of pushing the root loci toward the right-half plane.

2. Addition of zeros of G(s)H(s)

Example $G(s)H(s)=\dfrac{K}{s(s+4)}$

Add two zeros at s=-5±3j, we have

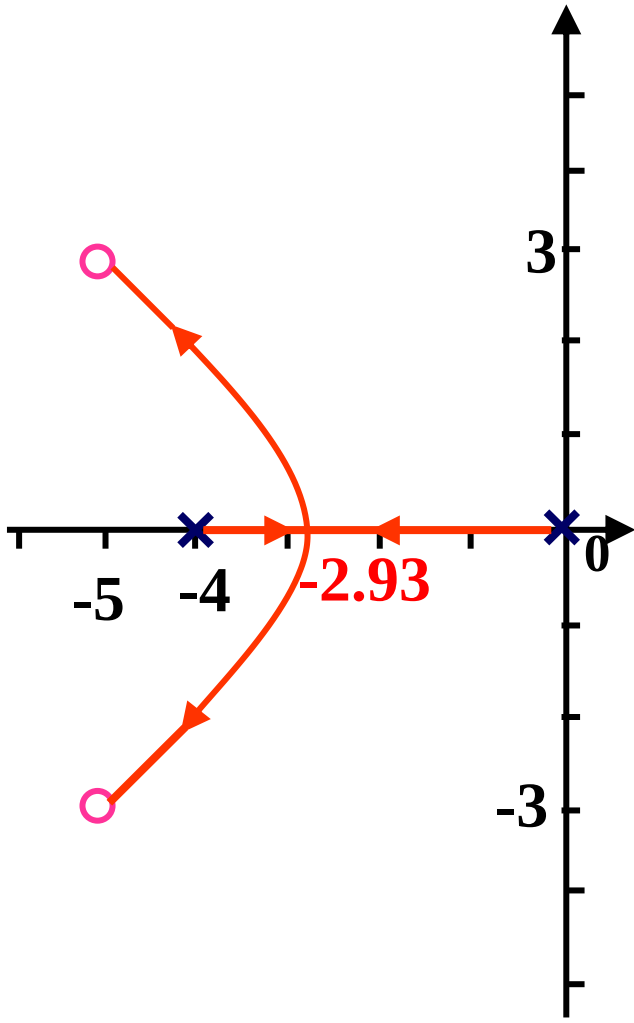
$$G(s)H(s)=\frac{K(s^2+10s+34)}{s(s+4)}$$

Poles: -4, 0 zero: -5±3j, n=2,m=2

RL on real-axis: [-4,0]

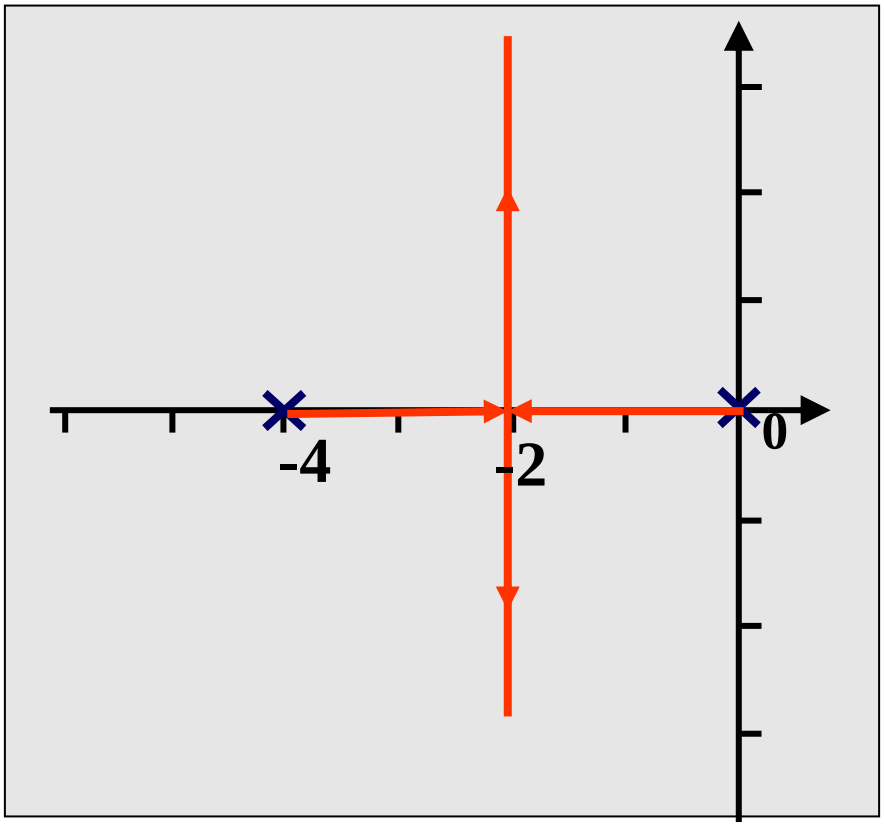
Breakaway point:

$$\frac{dG(s)H(s)}{ds}=0 \quad \Rightarrow \quad s=-2.93$$



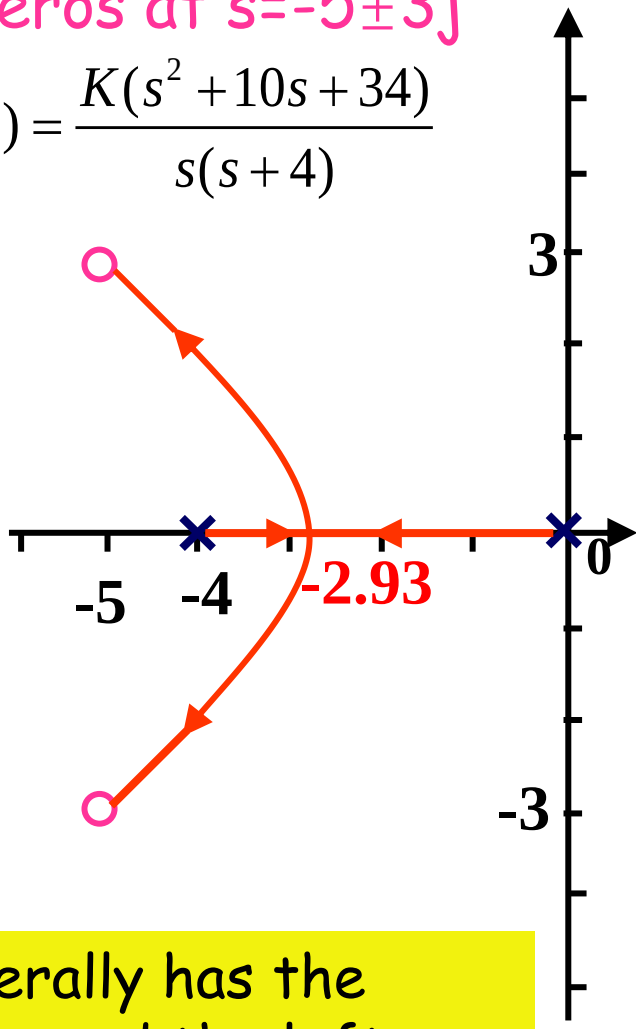
2. Addition of zeros of G(s)H(s)

Example $G(s)H(s) = \dfrac{K}{s(s+4)}$



Add two zeros at s=-5±3j

$$G(s)H(s) = \frac{K(s^2+10s+34)}{s(s+4)}$$

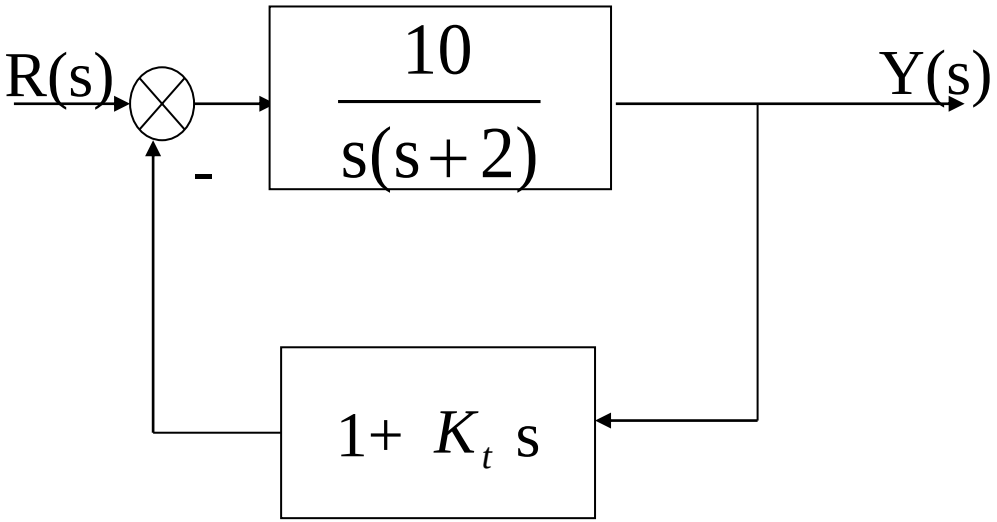


Adding left-half plane zeros to G(s)H(s) generally has the effect of moving and bending the root loci toward the left-half plane.

RL for the case of varying zeros of G(s)H(s)

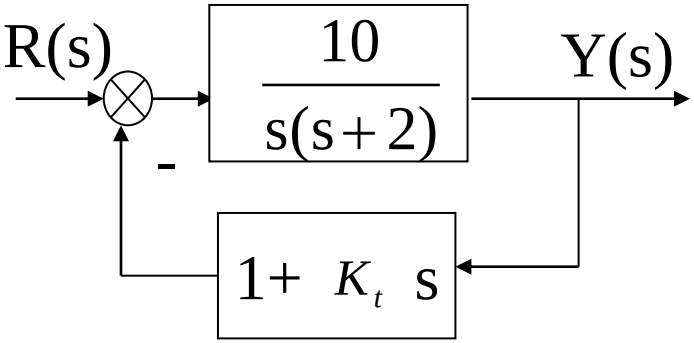
Example 1. Given a system's structure diagram, where the parameter K_t is speed feedback constant.

Problem: Draw RL when K_t varies from 0 to ∞.



The RL loop transfer function is

$$G(s)H(s) = \frac{10(1+K_t s)}{s(s+2)}$$



The closed-loop transfer function is:

$$T(s) = \frac{G(s)}{1+G(s)H(s)} = \frac{10}{s^2+2s+10+10K_t s}$$

The characteristic equation is: $s^2 + 2s + 10 + 10K_t s = 0$

It can be re-written as

$$1 + \frac{10K_t s}{s^2 + 2s + 10} = 0$$

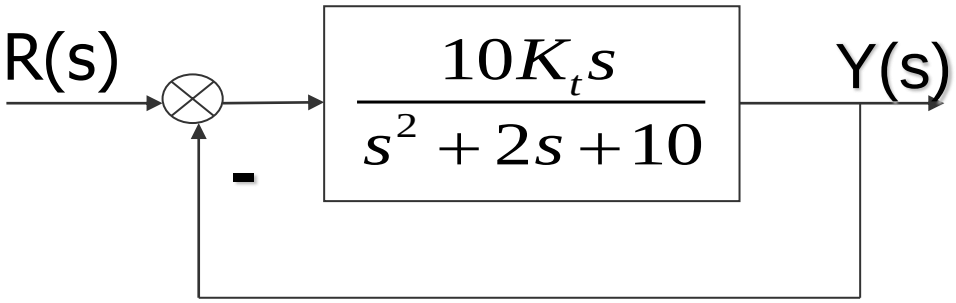
Define

$$G_1(s) = \frac{10K_t s}{s^2 + 2s + 10} = \frac{K_t^* s}{(s + 1 + j3)(s + 1 - j3)}$$

$$K_t^* = 10K_t$$

The $G_1(s)$ is the equivalent RL loop transfer function whose RL loop gain is $10K_t$

The equivalent system is shown as follows:



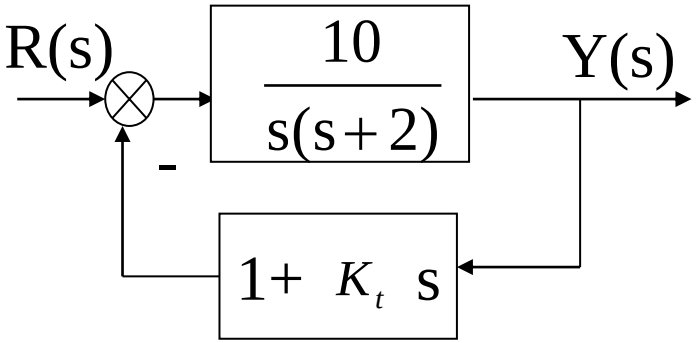
The closed-loop transfer function :

$$T(s)=\frac{\dfrac{10K_t s}{s^2+2s+10}}{1+\dfrac{10K_t s}{s^2+2s+10}}=\frac{10K_t s}{s^2+2s+10+10K_t s}$$

Note: The same denominator, but different numerator, compared with the original system.

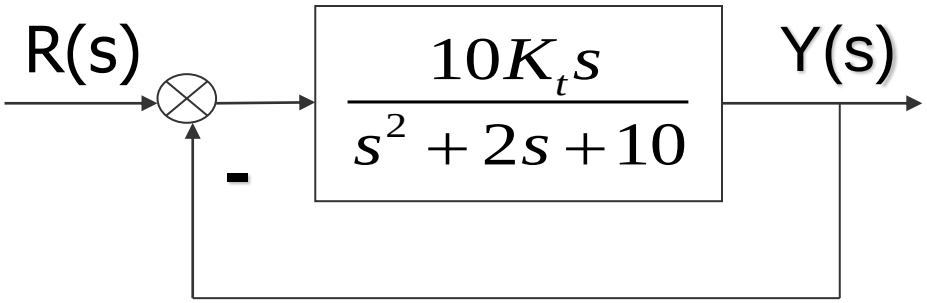
The original and equivalent systems

- Original system



$$T(s) = \frac{10}{s^2 + 2s + 10 + 10K_t s}$$

- Equivalent system



$$T(s) = \frac{10K_t s}{s^2 + 2s + 10 + 10K_t s}$$

The two systems have same closed loop characteristic equation (same denominator), but different closed loop zeros (different numerator).

For the equivalent system: 2 poles, 1 zeros
n=2, m=1

$$G_1(s) = \frac{K_t^* s}{s^2 + 2s + 10}$$

$$(K_t^* = 10K_t)$$

$$p_1 = -1 + j3, \quad p_2 = -1 - j3, \quad z_1 = 0$$

2 branches, one approaches zero, the other approaches infinity

There are RL on the real axis [-∞, 0]

The breakaway point d :

$$1/(d-p_1) + 1/(d-p_2) = 1/(d-z_1)$$

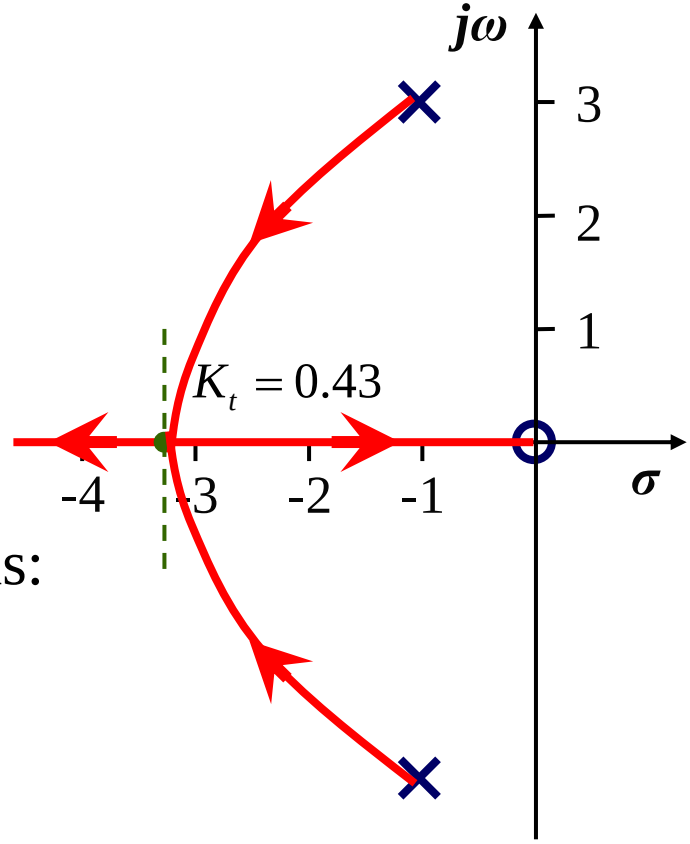
$$d_1 = -3.16, \quad d_2 = 3.16 \text{ (not within range)}$$

From $1 + G_1(s) = 0$, the corresponding gain K_t is:

$$K_t = -(s^2 + 2s + 10)/10s = 0.43$$

Angle of departure from p_1

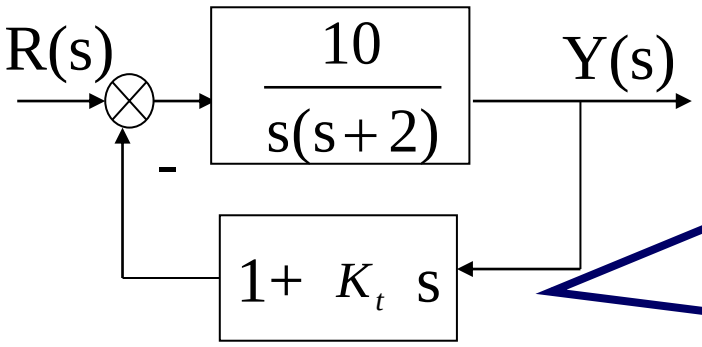
$$\theta_{p1} = 180^o + 108.43^o - 90^o = 198.43^o$$



Remark

With the root loci drown using the equivalent open-loop transfer function, we can only determine system closed-loop poles, we still need to determine the system closed-loop zeros to analyze system dynamic behavior.

Analysis of the influence of variation of K_t on the system



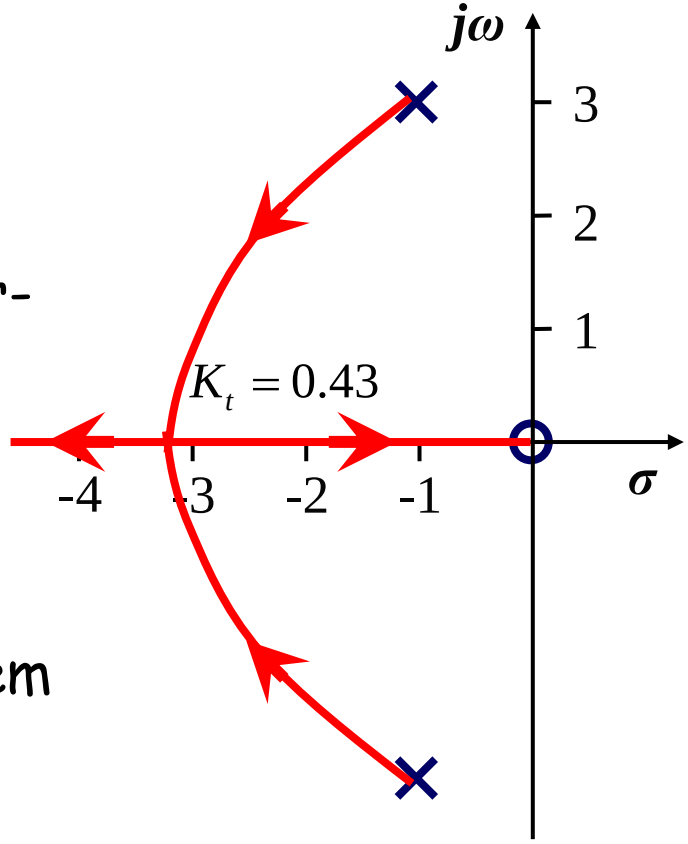
When K_t varies from 0 to infinity, a zero is added to the loop function G(s)H(s) and the zero moves from the negative infinity to the origin of s-plane.

When K_t is small, a pair of closed-loop complex-conjugate poles are close to the Im-axis, the overshoot of system step response is large and the oscillation is strong. When K_t is small, the system velocity feedback signal is weak and under-damped.

When K_t increases, the system damping is enhanced, the oscillation is weakened and the overshoot decreases, the system behavior is enhanced.

The breakaway point: $K_t = 0.43$, the system is critically damped.

When $K_t > 0.43$, the two closed-loop poles are negative and real, the system is over-damped.



Adding left-half plane zeros to open-loop transfer function generally has the effect of moving and bending the root loci toward the left-half s-plane.

RL for the case of varying poles of G(s)H(s)

Example 2: Consider a unity feedback system whose open-loop transfer function is

$$G(s)H(s) = \frac{K}{s(Ts+1)(\tau s+1)}$$

where K and T are given constants, while the parameter τ is to be determined.

Problem: Draw RL when τ varies from 0 to infinity.

$$G(s)H(s)=\frac{K}{s(Ts+1)(\tau s+1)}$$

The characteristic equation:

$1+G(s)H(s)=0$

$$\tau s^2(Ts+1)+s(Ts+1)+K=0 \quad \Rightarrow \quad 1+\frac{\tau s^2(Ts+1)}{s(Ts+1)+K}=0$$

The equivalent open loop transfer function $G_1(s)$:

$$G_1(s)=\frac{\tau s^2(Ts+1)}{s(Ts+1)+K}=\frac{\tau s^2(Ts+1)}{Ts^2+s+K}$$

Solving $Ts^2+s+K=0$ yields $s_{1,2}=\dfrac{-1\pm\sqrt{1-4TK}}{2T}$

Suppose that 4KT > 1, then two complex poles of G(s) are:

$$P_1=-\frac{1}{2T}+j\sqrt{\frac{K}{T}-(\frac{1}{2T})^2},\ P_2=-\frac{1}{2T}-j\sqrt{\frac{K}{T}-(\frac{1}{2T})^2}$$

$$G_1(s)=\frac{\tau s^2(Ts+1)}{(s-P_1)(s-P_2)}$$

$$P_1=-\frac{1}{2T}+j\sqrt{\frac{K}{T}-(\frac{1}{2T})^2},\ P_2=-\frac{1}{2T}-j\sqrt{\frac{K}{T}-(\frac{1}{2T})^2}$$

2 poles, 3 zeros $Z_1=Z_2=0$, $Z_3=-1/T$

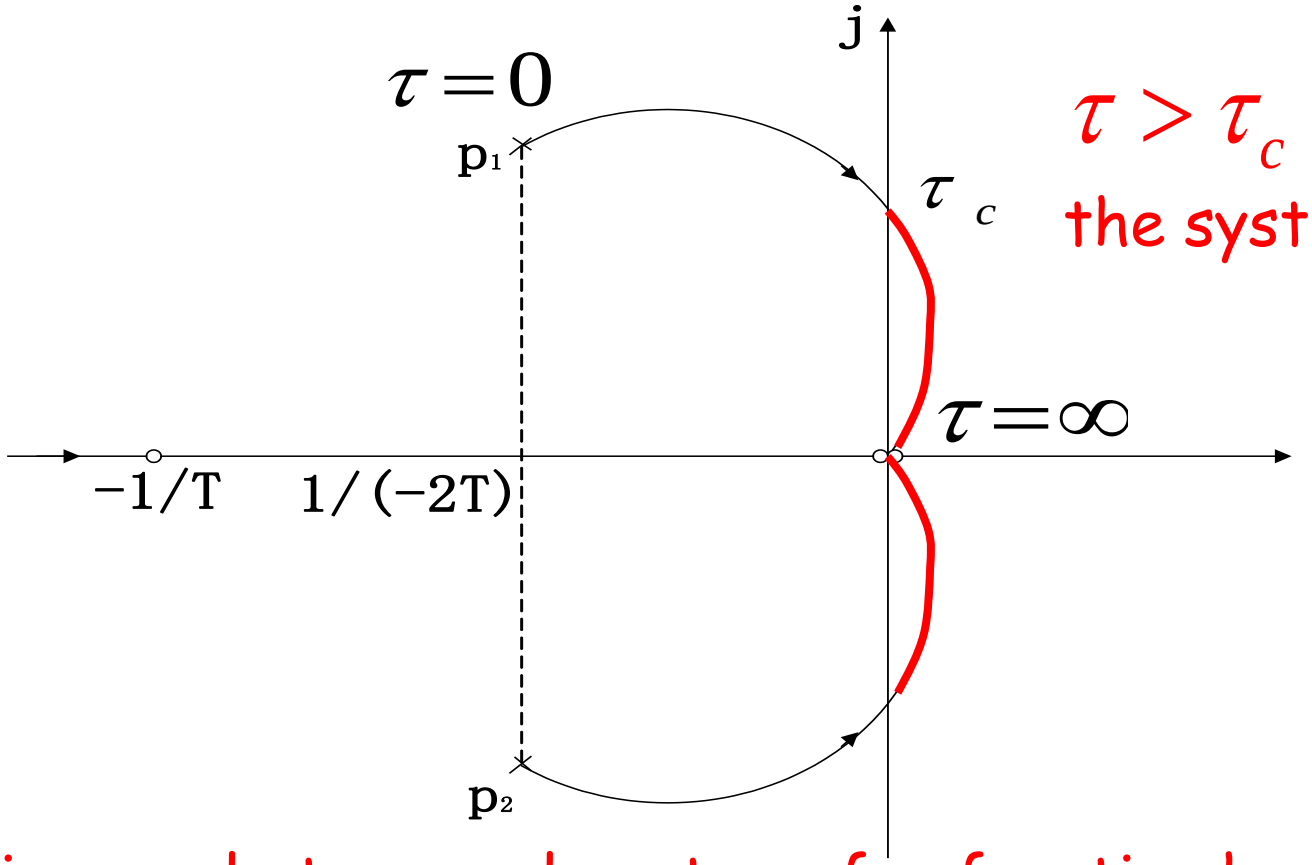
So, number of branches of the RL is equal to 3.

According to rule 8, the intersection point with the imaginary axis is determined by $1+G_1(j\omega)=0$, that is

$$1+\frac{\tau_c(j\omega)^2(j\omega T+1)}{j\omega(j\omega T+1)+K}=0\Rightarrow\begin{cases}K-\omega^2T-\tau_c\omega^2=0\\ \omega-\tau_cT\omega^3=0\end{cases}$$

It follows that $\omega=\frac{\sqrt{KT-1}}{T},\tau_c=\frac{T}{KT-1}$

RL when τ varies from 0 to infinity is shown as follows:

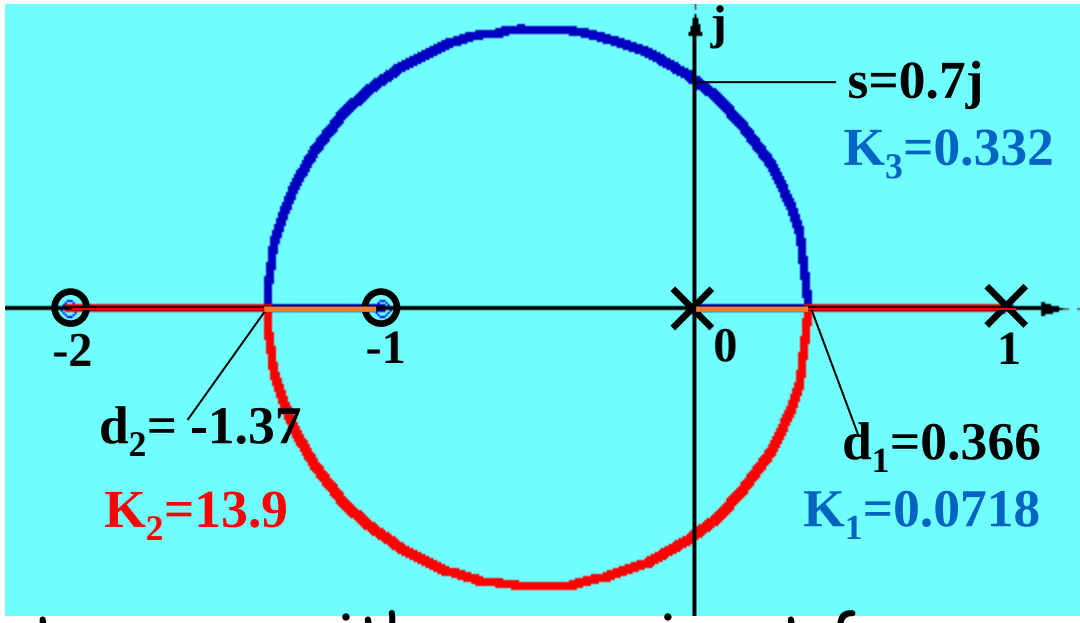


$\tau > \tau_c$ the system is unstable.

Adding a pole to open-loop transfer function has
The effect of pushing the RL toward the right-half plane

Example

Consider the RL of a unity-feedback system



1 Determine the closed-loop TF when its CE has multi-order roots.

2 Determine the steady-state error with a ramp input for an underdamped closed-loop system

$$G(s)=\frac{K(s+1)(s+2)}{s(s-1)}$$

Solution. 1. RL loop function:

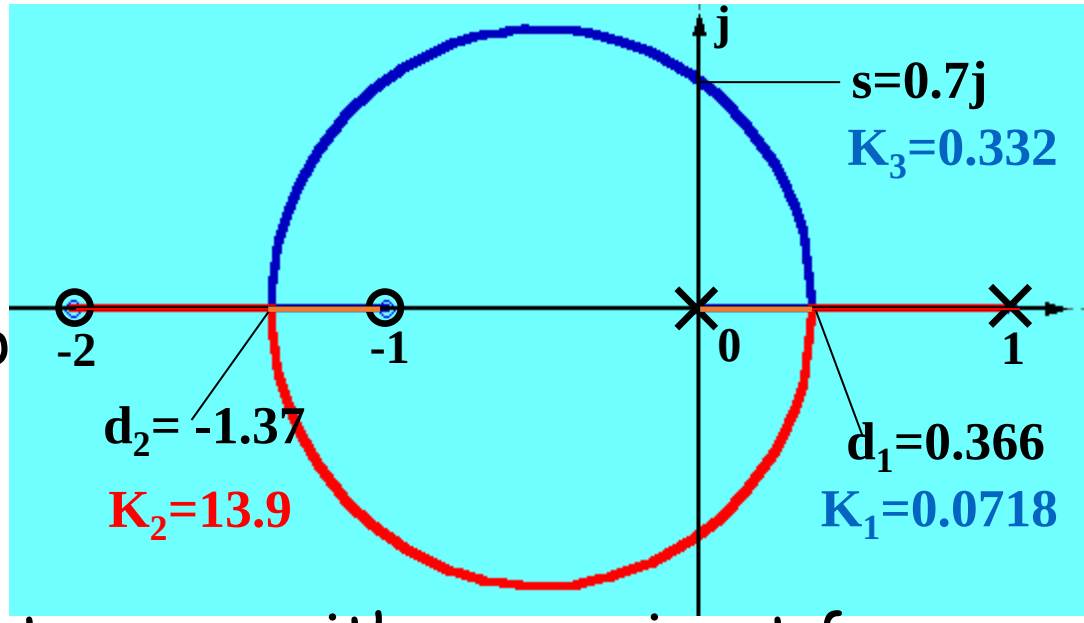
The CE of closed-loop system has multi-order roots at two breakaway points which are actually two closed-loop roots

$$d_1=0.366,\ K_1=0.0718 \qquad T(s)=\frac{G(s)}{1+G(s)}=\frac{0.067(s+1)(s+2)}{(s-0.366)^2}$$

$$d_2=-1.37,\ K_2=13.9 \qquad T(s)=\frac{G(s)}{1+G(s)}=\frac{0.933(s+1)(s+2)}{(s+1.37)^2}$$

Example

Consider the RL of a unity-feedback system



1 Determine the closed-loop TF when its CE has multi-order roots.

2 Determine the steady-state error with a ramp input for an underdamped closed-loop system

Solution. 2. RL loop function: $G(s)=\dfrac{K(s+1)(s+2)}{s(s-1)}$ Type-1 system:

For a ramp input, the velocity error constant: $K_v=\lim_{s\to 0} sG(s)=-2K$

Closed-loop system is underdamped: $0.332<K<13.9$

The steady-state error: $e_{ss}=\dfrac{1}{K_v}$

$$-1.5<e_{ss}<-0.036$$

Pole -Zero Cancellations

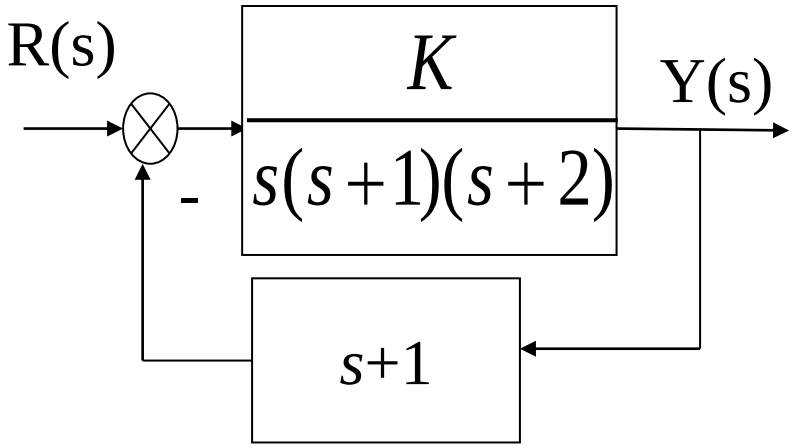
Pole –Zero Cancellations

If the denominator of G(s) and the numerator of H(s) have common factors (or vise versa), then the open-loop poles and zeros will cancel each other.

This will reduce the degree of the CE.

However, the cancelled pole (zero) of G(s) will continue to be a pole (zero) of the closed-loop system.

Example



$$\Rightarrow \quad G(s)H(s) = \frac{K}{s(s+2)}$$

The reduced CE is $1 + G(s)H(s) = 0 \Rightarrow s(s+2) + K = 0$

Pole –Zero Cancellations

We sketch the RL based on the reduced equation.

However, the actual closed-loop TF is

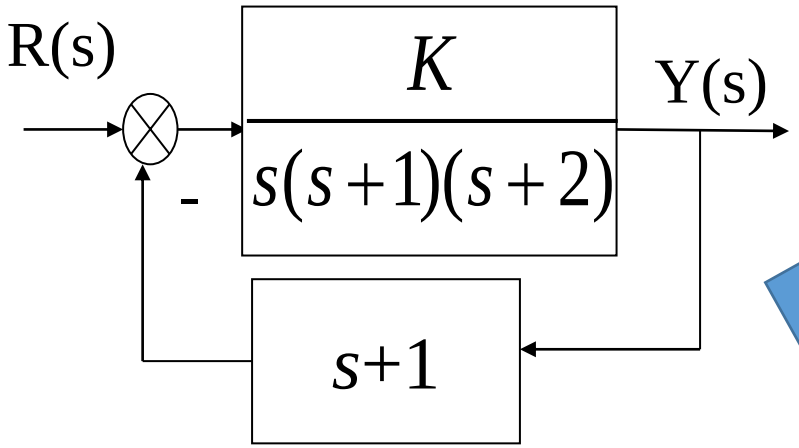
$$T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{K}{(s+1)\left[s(s+2)+K\right]}$$

$\Rightarrow$ $s = -1$ is still a pole of the CL TF

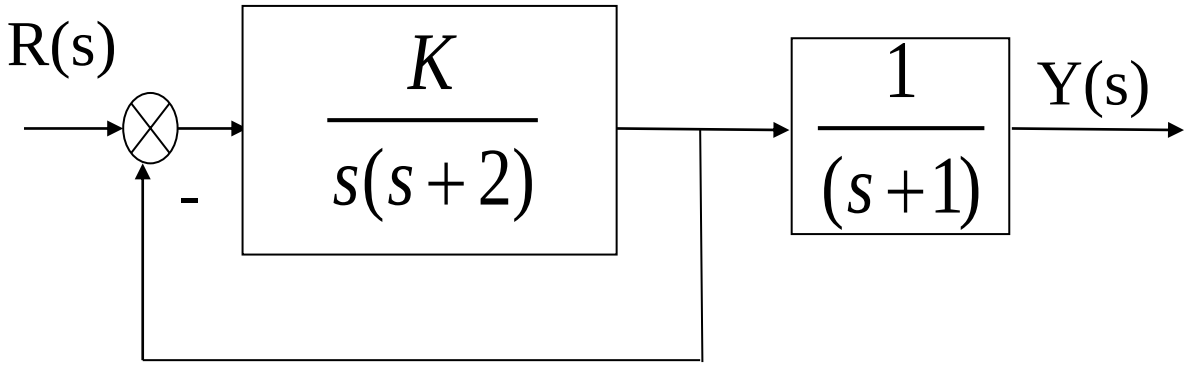
The pole s = -1 does not change with K.

Therefore, once the RL is drawn for the reduced CE, the cancelled pole must be added to the root locus in order to obtain a complete picture.

Pole –Zero Cancellations



The two systems have the same CL TF.



The two systems are equivalent from input-output point of view.

Parameter Design by Root Locus Technique

Originally the root locus method was developed to determine the loci of roots of the characteristic equation as the system gain, K, varies from 0 to ∞.

However, as we have seen, the effect of other system parameters may be readily investigated by using the root locus technique.

For example, a third-order equation of interest might be

$$s^3 + (3+\alpha)s^2 + 3s + 6 = 0$$

To ascertain the effect of the parameter α, we isolate the parameter and rewrite the equation in root locus form

$$1 + \frac{\alpha s^2}{s^3 + 3s^2 + 3s + 6} = 0$$

If there are more than one parameters we want to investigate, we must repeat the root locus approach for each parameter.

For example, a third-order characteristic equation with α and β as parameters

$$s^3 + s^2 + \beta s + \alpha = 0$$

The effect of varying β from 0 to ∞ is determined from the root locus equation

$$1 + \frac{\beta s}{s^3 + s^2 + \alpha} = 0$$

The denominator of the above RL equation is the characteristic equation of the system with $\beta = 0$.

Therefore, we must first evaluate the effect of varying α from 0 to ∞ by utilizing

$$s^3 + s^2 + \alpha = 0$$

Rewritten as RL form:

$$1 + \frac{\alpha}{s^2(s+1)} = 0$$

Then upon evaluating the effect of α, a value of α is selected and used to evaluate the effect of β.

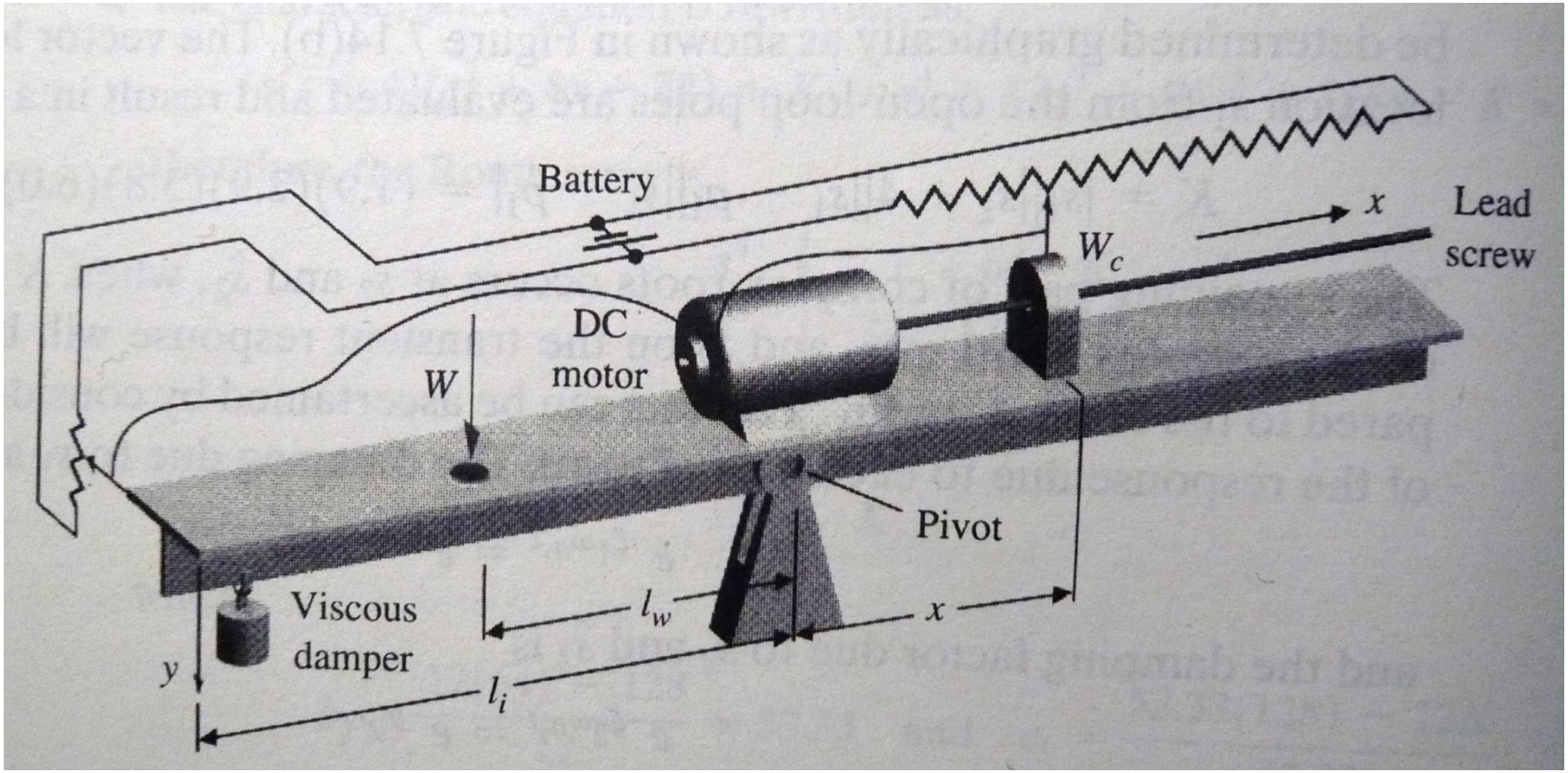
When more than one parameters varies continuously from 0 to ∞, the family of root loci are referred to as the root contours (RC).

Example

$$s^3 + 3s^2 + 2s + \beta s + \alpha = 0$$

An example of a control system analysis and design utilizing Root Locus technique

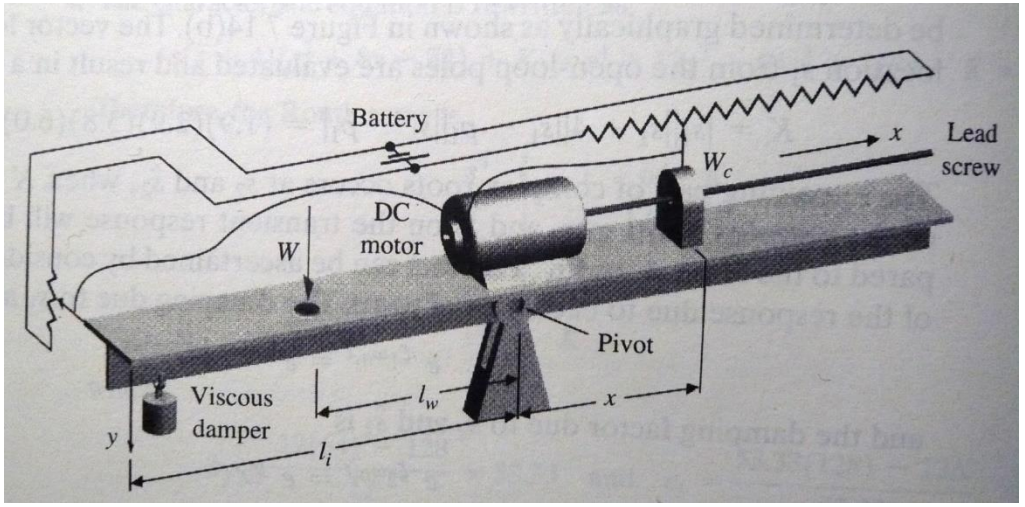
An automatic self-balancing scale



We desire to accomplish the following:

- Select the parameters of the system
- Determine the specifications
- Obtain a model to represent the system
- Design the gain K using root locus
- Determine the dominant mode of response

Parameters



Counterweight: $W_c = 2N$

Inertia of the beam: $I = 0.05 kgm^2$

Length of the weight to the pivot: $l_w = 5cm$

Length of the beam to the viscous damper: $l_i = 20cm$

Damping constant of the viscous damper: $b = 10\sqrt{3} kg/m/s$

Lead screw gain: $K_s = \frac{1}{4000\pi} m/rad$

Input potentiometer gain: $K_i = 4800V/m$

Feedback potentiometer gain: $K_f = 400V/m$

Specifications

- A rapid and accurate response resulting in a small steady-state weight measurement error is desired.

Specifications	
Steady-state error (for a step input)	$K_p=\infty$, $e_{ss=0}$
Underdamped response	$\zeta=0.5$
Settling time (error band $\Delta<2\%$)	$t_s<2sec$

Mathematical model

For small deviations from balance, the deviation angle is

$$\theta \simeq \frac{y}{l_i}$$

The motion of the beam about the pivot is represented by the torque equation:

$$I\frac{d^2\theta}{dt^2} = \Sigma torques$$

$$I\frac{d^2\theta}{dt^2} = l_w W - xW_c - l_i^2 b\frac{d\theta}{dt}$$

Mathematical model

The input voltage to the motor is

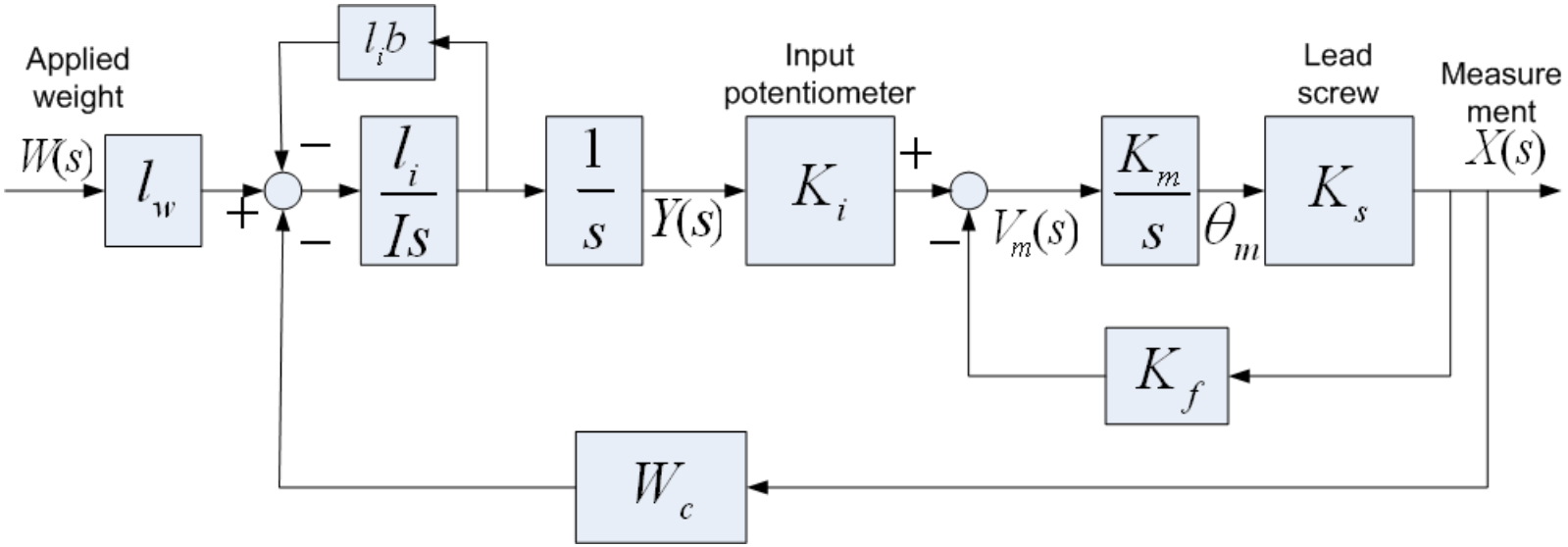
$$v_m(t) = K_i y - K_f x$$

The transfer function of the motor is

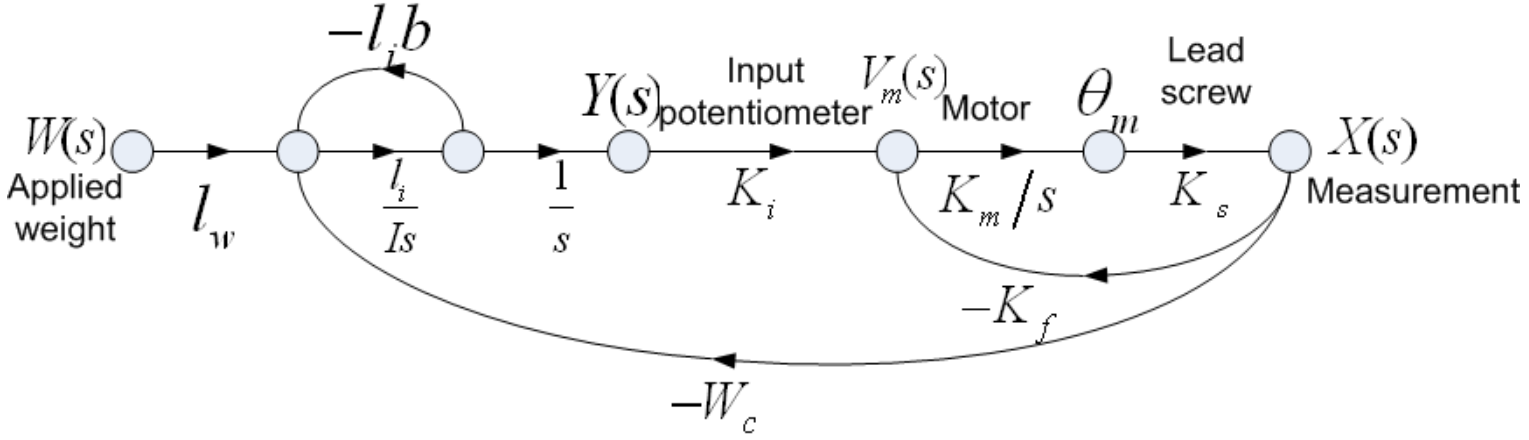
$$\frac{\theta_m(s)}{V_m(s)} = \frac{K_m}{s(\tau s + 1)} \simeq \frac{K_m}{s}$$

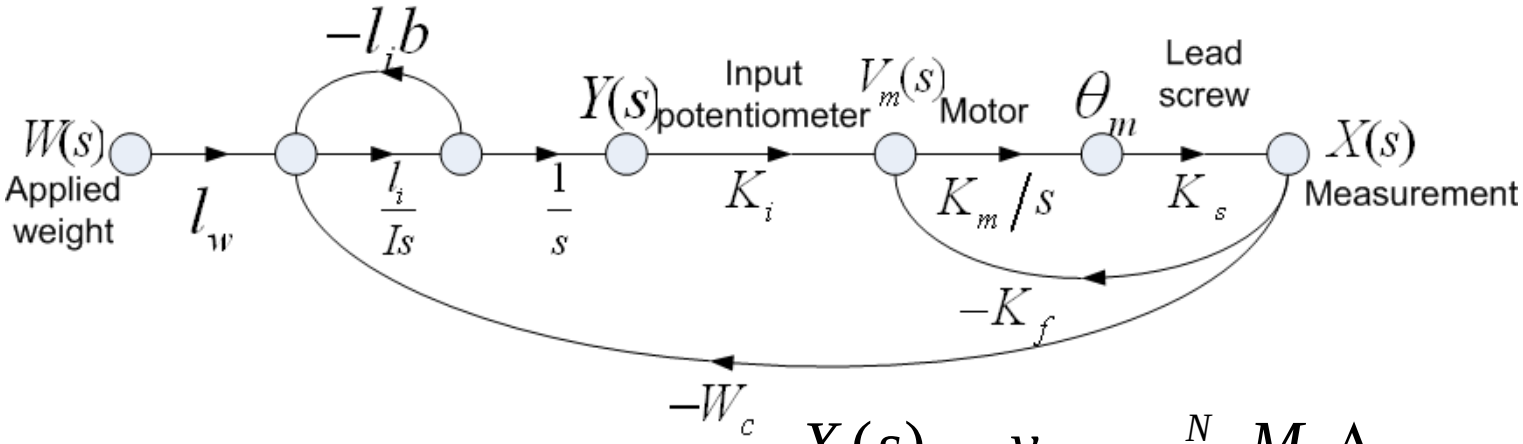
where τ is negligible with respect to the time constants of the overall system and θ_m is the output shaft rotation.

Block diagram



Signal-flow





Using Mason's formula: $\dfrac{X(s)}{W(s)} = \dfrac{y_{out}}{y_{in}} = \sum_{k=1}^{N} \dfrac{M_k \Delta_k}{\Delta}$

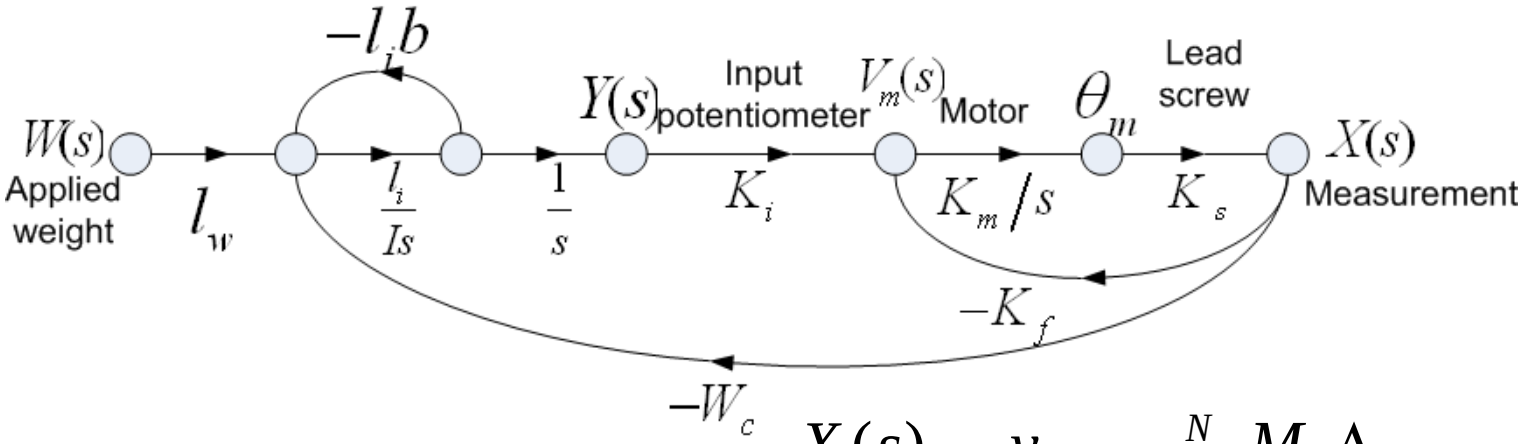
Forward path: $M_1 = l_w l_i K_i K_m K_s / Is^3, \quad \Delta_1 = 1$

Loop path: $L_{11} = l_i^2 b / Is, \quad L_{21} = K_m K_s K_f / s$

$L_{31} = l_i K_m K_s K_f W_c / Is^3$

Nontouching loops: $L_{12} = l_i^2 b K_m K_s K_f / Is^2$

$\Delta = 1 - (L_{11} + L_{21} + L_{31}) + L_{12}$



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The closed-loop transfer function:

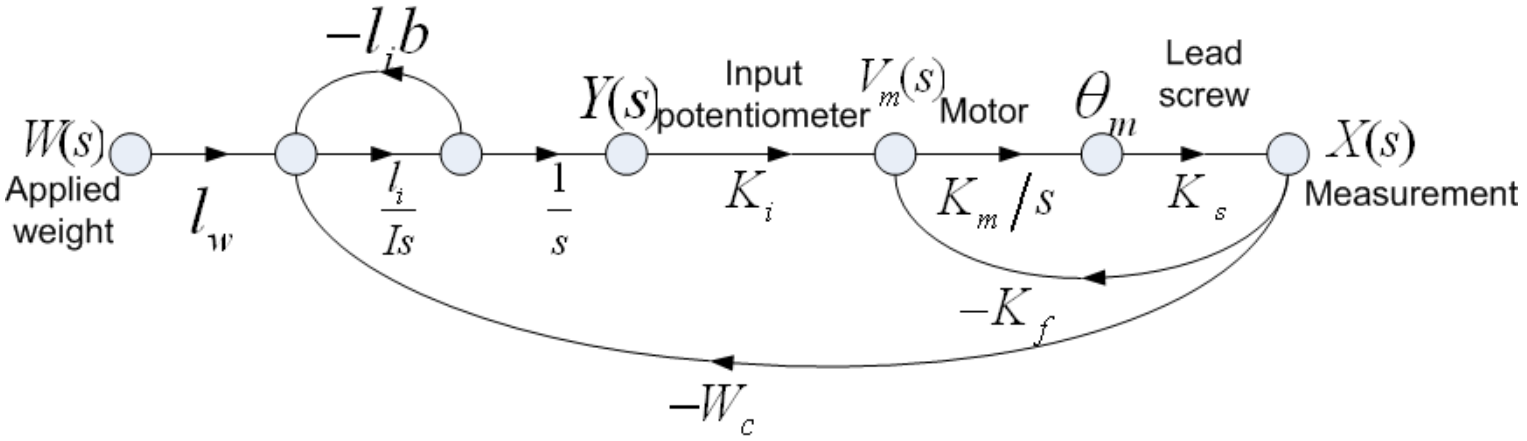
$$T(s) = \frac{X(s)}{W(s)} = \frac{l_w l_i K_i K_m K_s}{s(Is + l_i^2 b)(s + K_m K_s K_f) + W_c K_m K_s K_i l_i}$$

The steady-state gain of the system is:

$$\lim_{s \to 0} \left(\frac{X(s)}{W(s)} \right) = \frac{l_w l_i K_i K_m K_s}{W_c K_m K_s K_i l_i} = \frac{l_w}{W_c} = \frac{0.05}{2} = 0.025 m / kg$$

Characteristic equation:

$$s(Is + l_i^2 b)(s + K_m K_s K_f) + W_c K_m K_s K_i l_i = 0$$

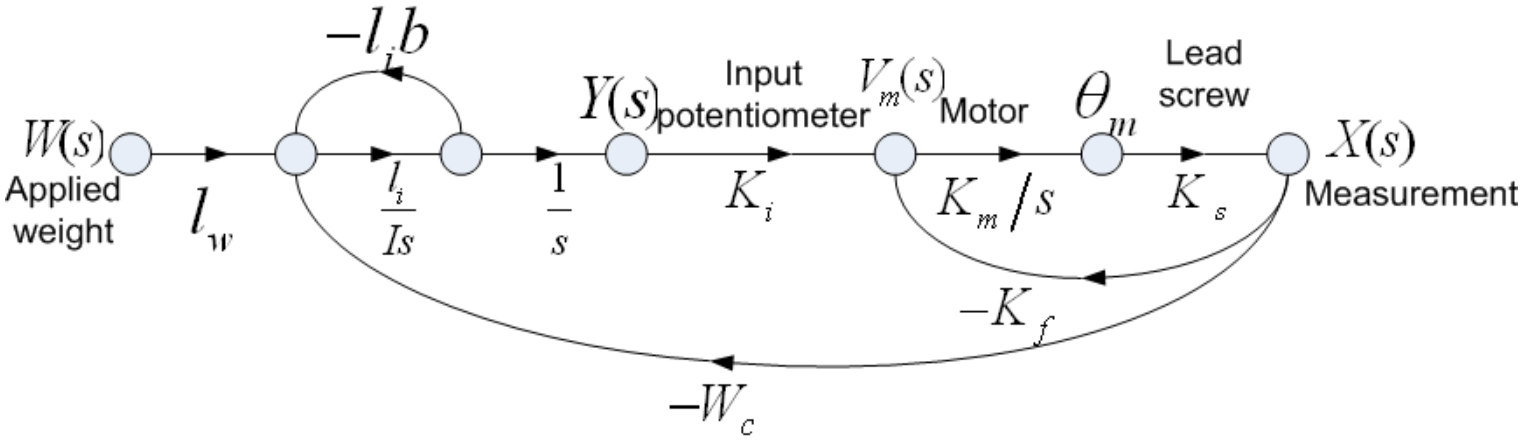


Substituting the parameter values into the CE yields:

$$s\left(s+8\sqrt{3}\right)\left(s+\frac{K_m}{10\pi}\right)+\frac{96K_m}{10\pi}=0$$

To rewrite it into root locus form, first isolate K_m as follows:

$$s^2\left(s+8\sqrt{3}\right)+s\left(s+8\sqrt{3}\right)\frac{K_m}{10\pi}+\frac{96K_m}{10\pi}=0$$



Rewrite it into root locus form:

$$1+KG(s)H(s)$$

$$=1+\frac{(K_m/10\pi)(s+6.93+j6.93)(s+6.93-j6.93)}{s^2(s+8\sqrt{3})}=0$$

$$K=K_m/10\pi$$

$$G(s)H(s)=\frac{K(s+6.93+j6.93)(s+6.93-j6.93)}{s^2(s+13.8564)}$$

Closed-loop CE:

$$1+K\frac{(s+6.93+j6.93)(s+6.93-j6.93)}{s^2(s+13.8564)}=0$$

We draw the root locus as K varies from 0 to ∞

With MATLAB:

```
>>z=[-6.93-6.93i, -6.93+6.93i];
>>p=[0 0 -13.8564];
>>k=1;
>>sys=zpk(z,p,k);
>>rlocus(sys)
```

The dominant closed-loop poles can be placed at:

$$\zeta=0.5, \quad K=25.3$$

To achieve this gain: $K_m=10\pi\times K=795\frac{rad/s}{volt}$

Let us observe the time-domain step response of the closed-loop system with the designed Km in MATLAB to see if it satisfied the desired specifications.

Specifications	
Steady-state error (for a step input)	$K_p=\infty$, $e_{ss=0}$
Underdamped response	$\zeta=0.5$
Settling time (error band $\Delta<2\%$)	$t_s<2$sec